

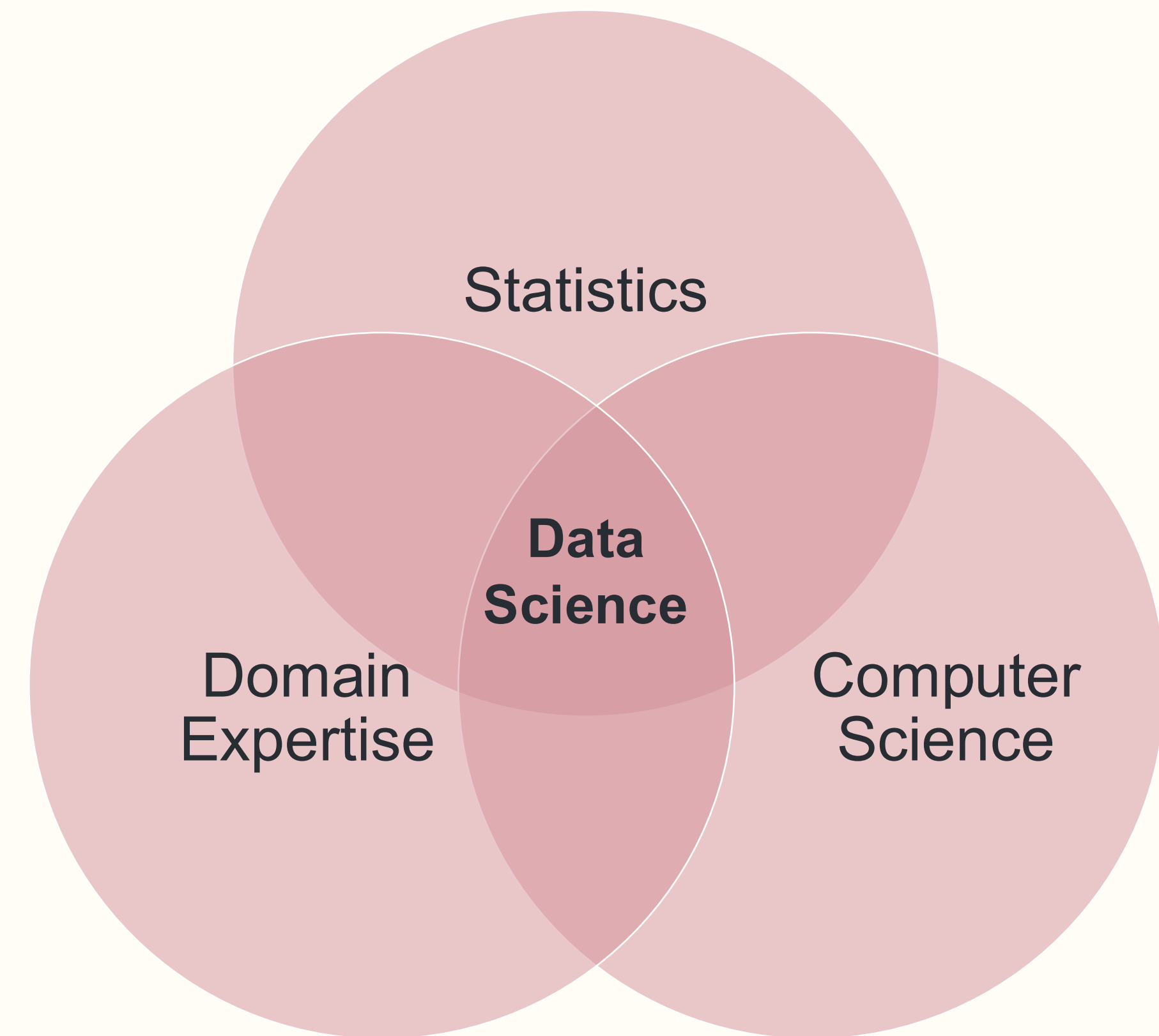
Lecture 1: Introduction to course, R, RStudio, and R Markdown

Heather Mattie, PhD

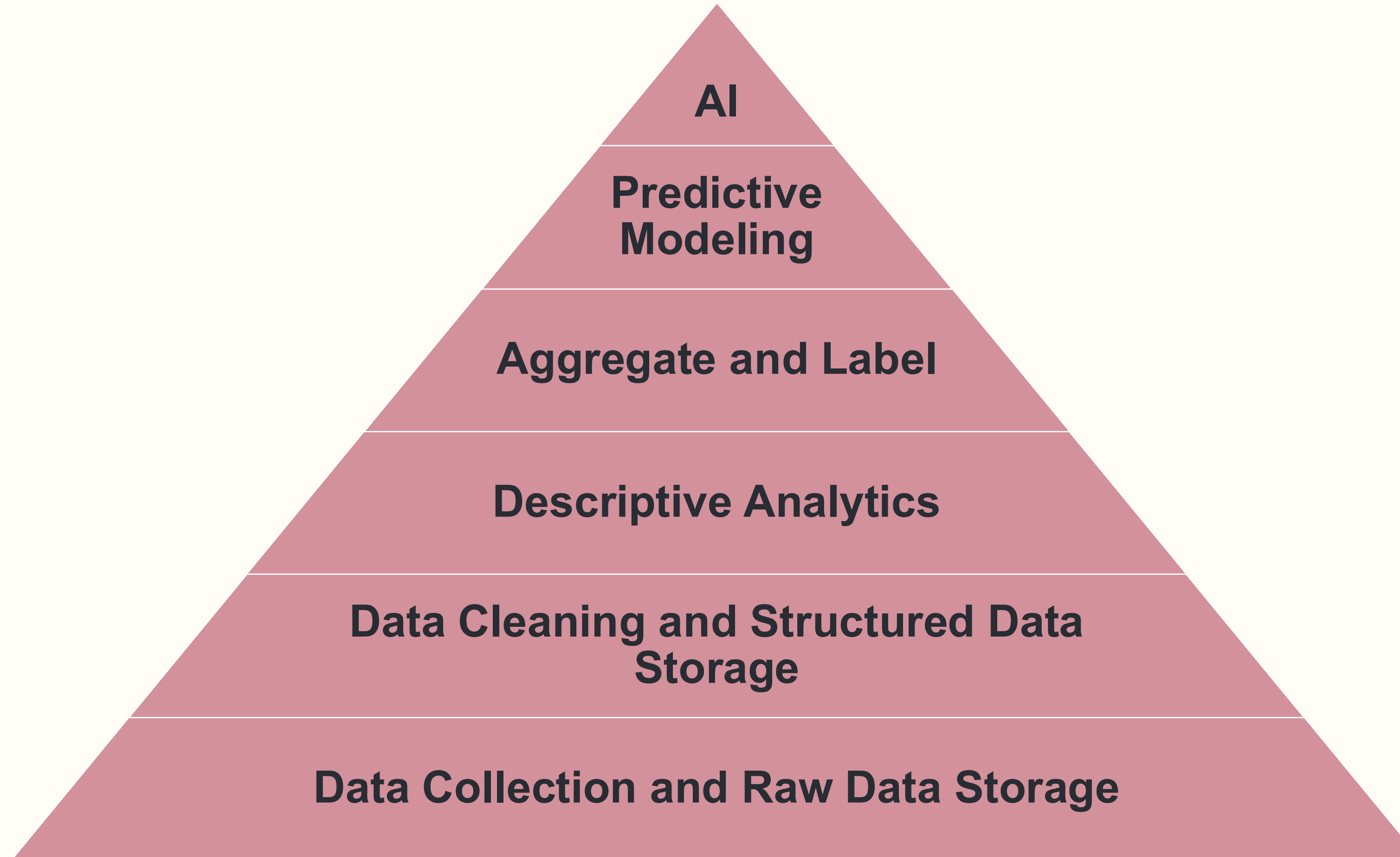
September 1, 2026

What is Data Science?

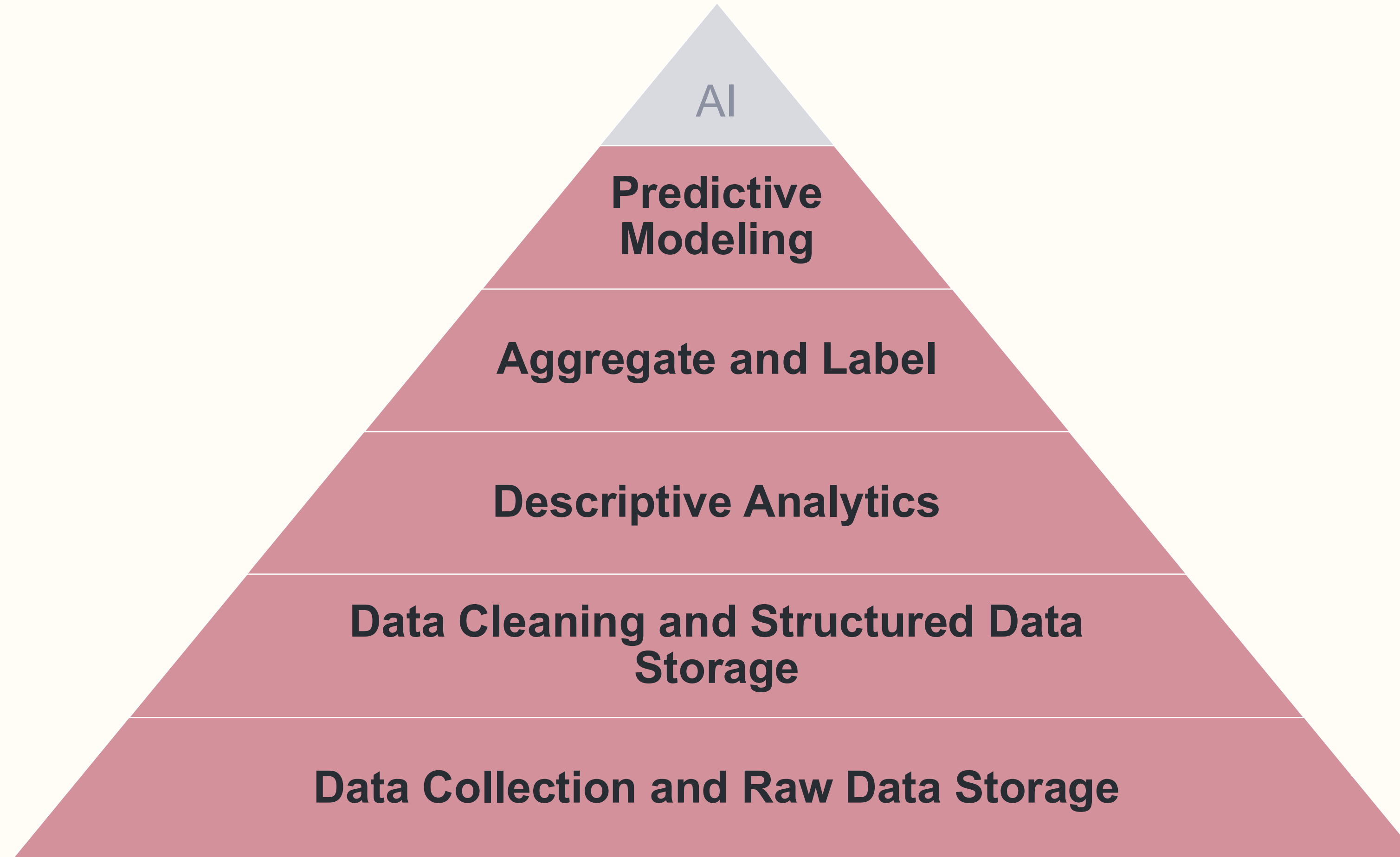
Data Science is a **multidisciplinary** field that uses scientific methods to **extract knowledge** and **insights** from **data**



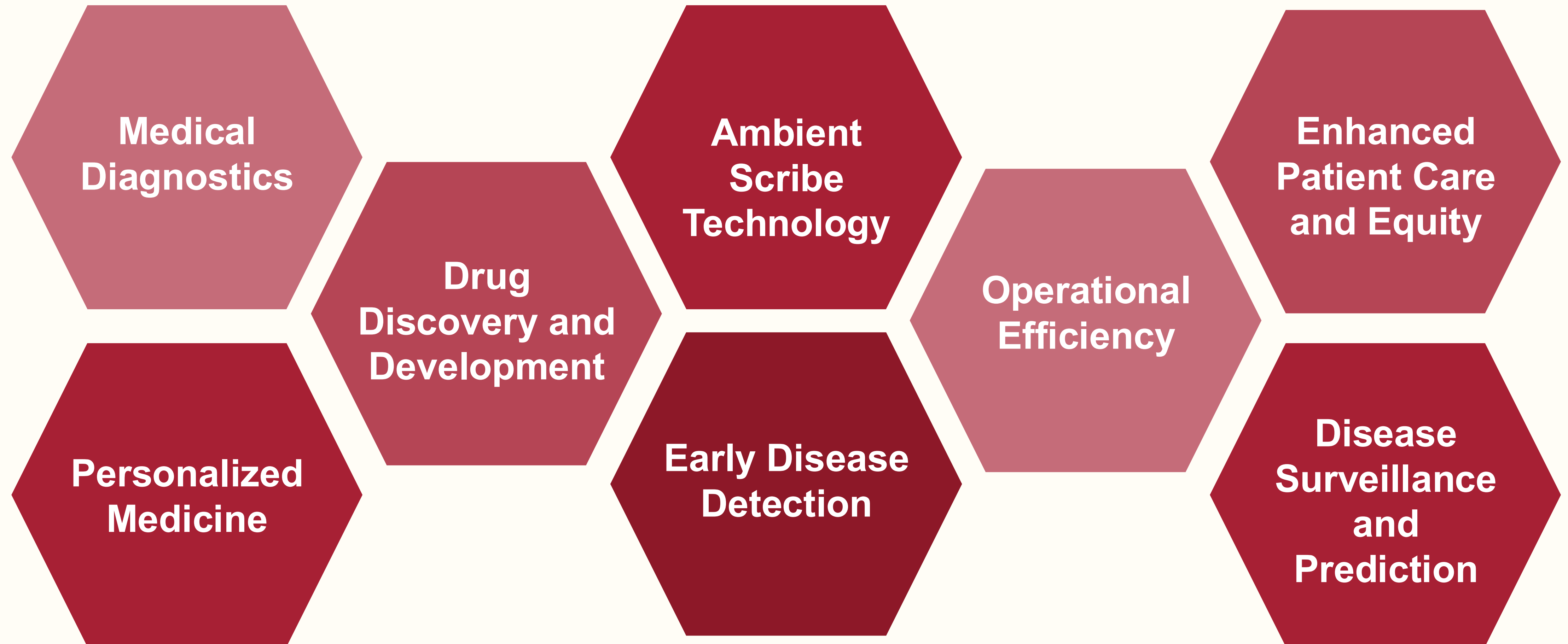
Data Science Hierarchy of Needs



Data Science Hierarchy of Needs



Data Science in Public Health – Successes



Data Science in Public Health – Pitfalls

Biased Algorithms

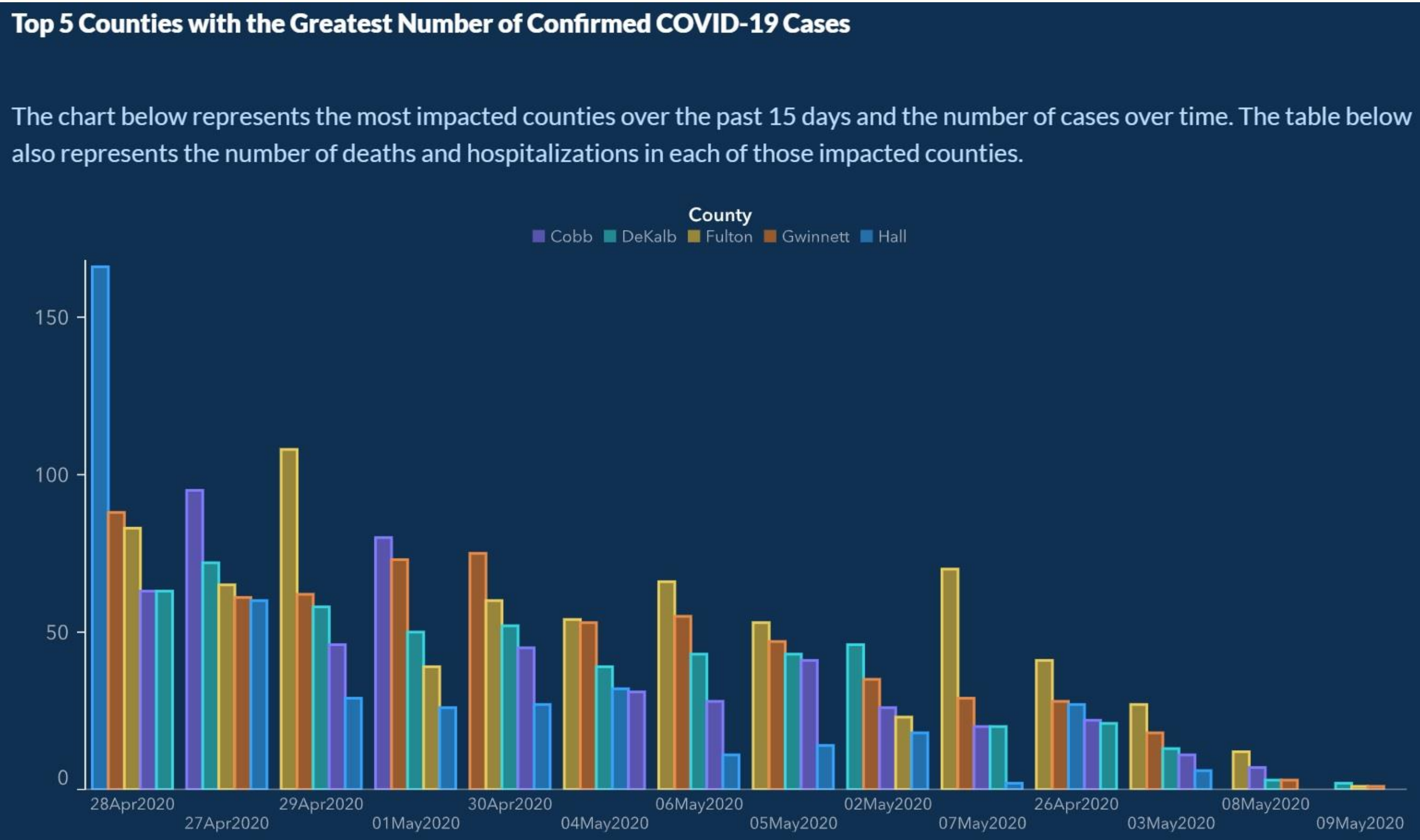
Data Privacy and Security Risks

Lack of Transparency

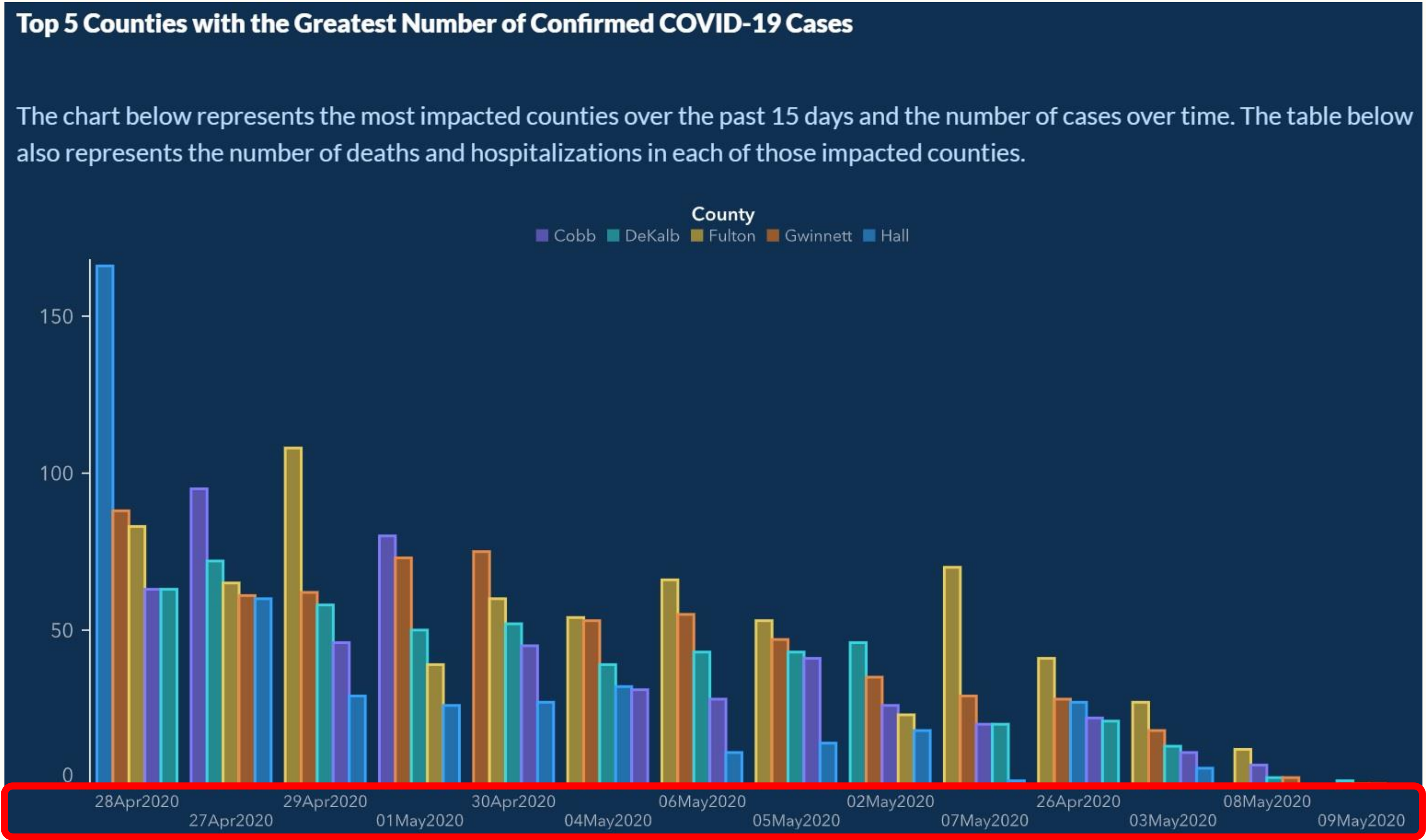
AI Regulation Landscape

Mis- and Disinformation

Misinformation Example



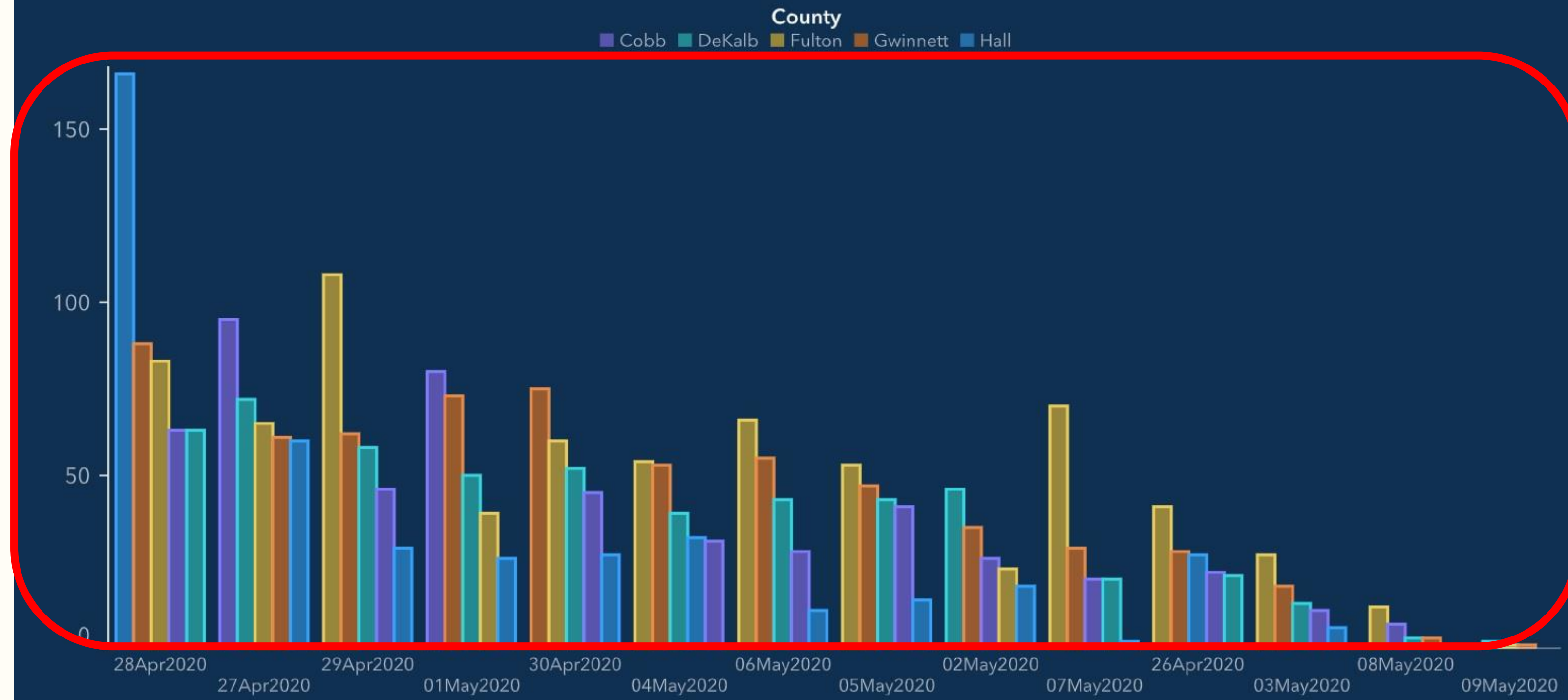
Misinformation Example



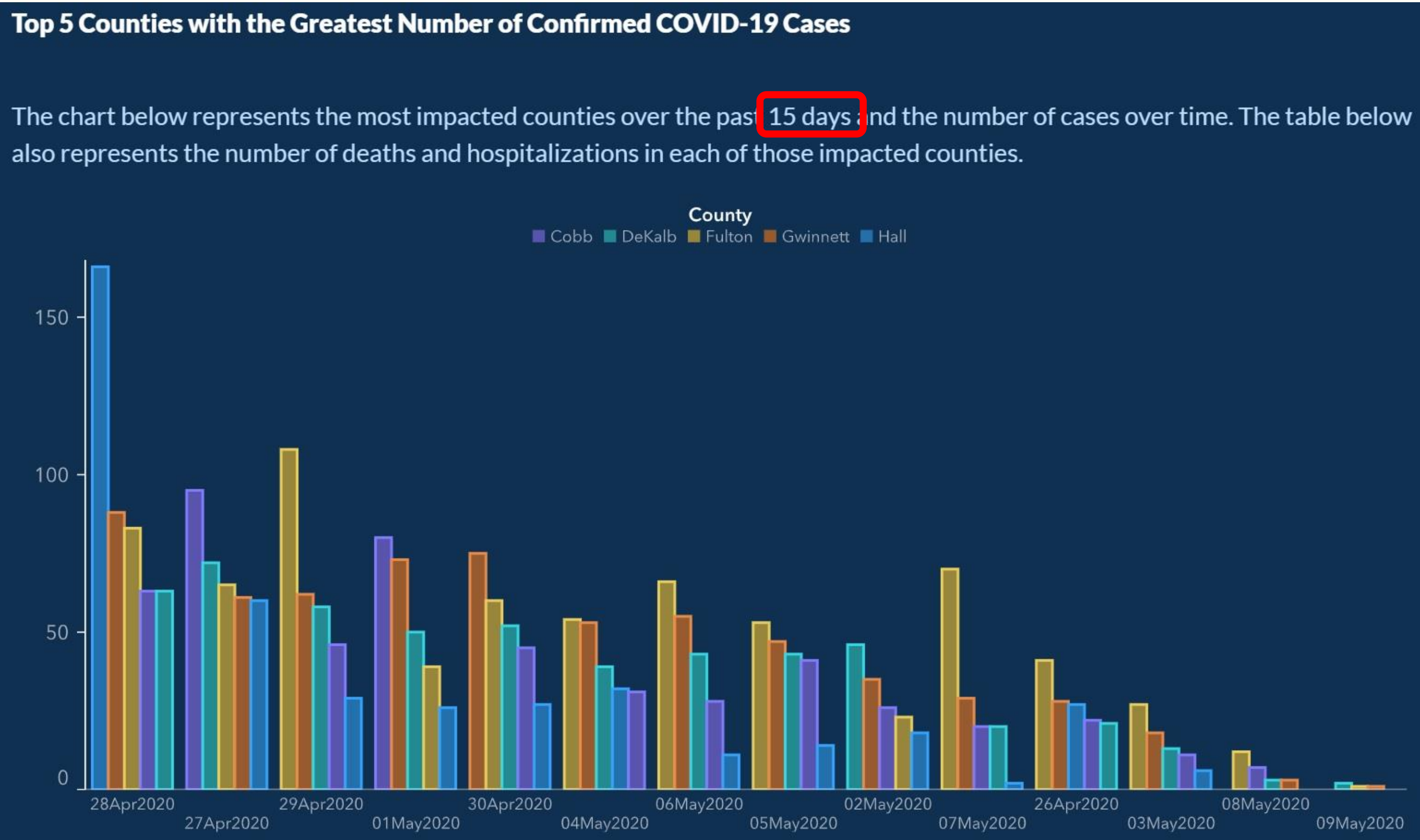
Misinformation Example

Top 5 Counties with the Greatest Number of Confirmed COVID-19 Cases

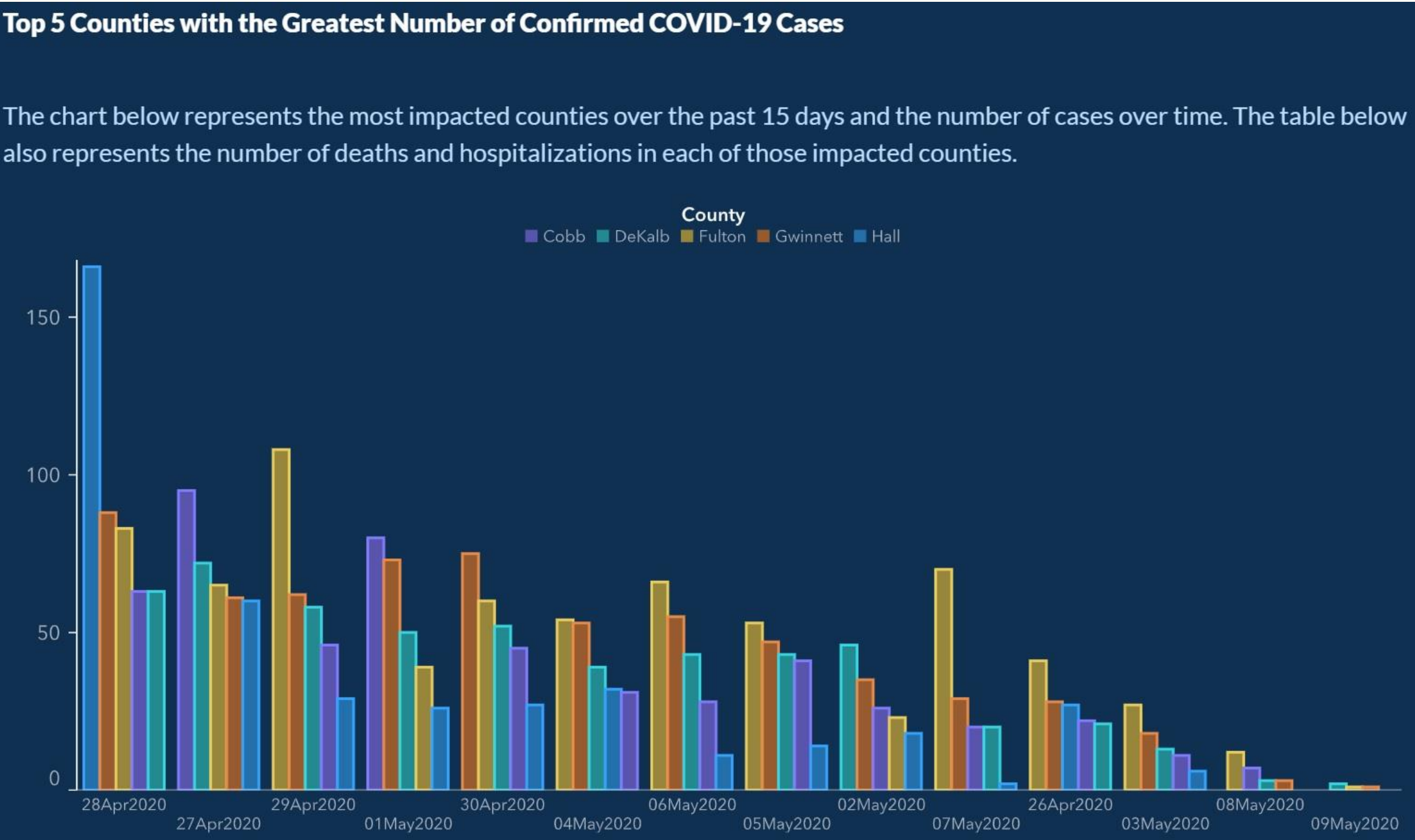
The chart below represents the most impacted counties over the past 15 days and the number of cases over time. The table below also represents the number of deaths and hospitalizations in each of those impacted counties.



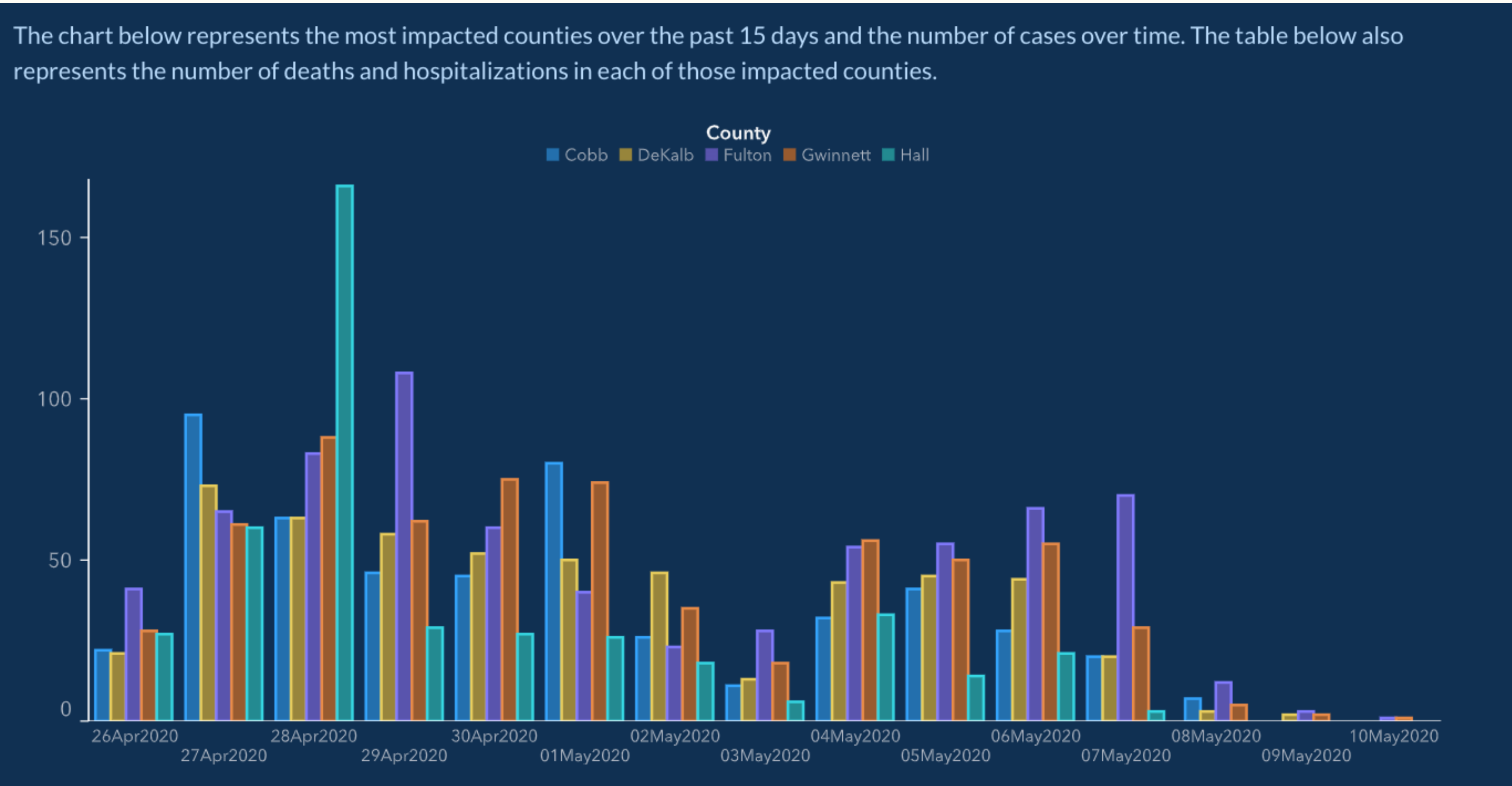
Misinformation Example



Misinformation Example



Corrected version



As data
[scientists, analysts, creators, consumers],
we all play a role in **understanding** and **confronting**
ethics, equity, and bias in data, algorithms, clinical
AI, and healthcare

The Data Science Pipeline



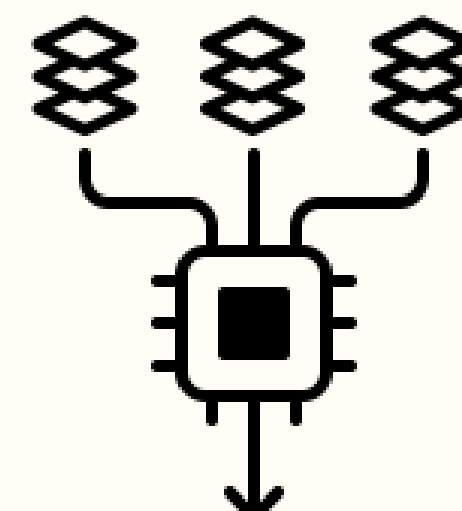
**Collecting
and
importing
(loading)
the data**



**Processing
(cleaning,
wrangling)
the data**



**Visualizing
and
summarizing
the data**



**Building
models
(statistical
and ML)**



**Interpretation
and
communication
of results**

Who is helping you learn data science?

Instructor

Heather Mattie
Lecturer on Biostatistics
hemattie@hsph.harvard.edu



Teaching Fellows

Stephanie Armbruster
sarmbruster@g.harvard.edu

Monica Robles Fontan
mroblesfontan@g.harvard.edu

Kimberly Greco
kimberly.greco@g.harvard.edu

Office Hours and Lab

Office Hours

Day	Time	Staff	Location
Thursday	12-1 pm	Heather	Building 1, 4 th floor, room 421A
Wednesday	10:30-11:30 am	Stephanie	Building 2, 4 th floor, room 426
Friday	11:30-12:30 pm	Monica	Building 2, 4 th floor, room 426
Monday	1-2 pm	Kimberly	Zoom

Lab

Day	Time	Location
Friday	9:45-11:15 am	Zoom

***Lab will not be held every week**



Office Hours and Lab

- Heather's office hour this week will be Thursday, 2-2:30pm
- The TFs will not hold office hours this week
- There is no lab this week

Grading

Homework

- 5 assignments
- 60% of final grade
- You are welcome to discuss the course material and homework questions with others, but the work you turn in must be your own. Be sure to cite any sources you use, including generative AI.

Take-home Midterm

- 15% of final grade
- Questions that require writing code and short answers
- You are not allowed to work on or discuss this assignment with other students or use generative AI

Grading

Final Project

- 25% of final grade
- Will work in teams of 4-5 people
- Will choose 1 of 3 possible projects

Pass/Fail Threshold

- If taking course pass/fail, must earn final grade of 70% or more to pass

Auditing

- Auditors do not need to submit any assignments

Generative AI Policy

As a tool, generative AI can provide valuable assistance in understanding concepts, troubleshooting issues, or even offering general guidance (use discretion!). However, to ensure true learning in an introduction to data science course, you should not rely on generative AI to complete your homework/midterm/final project. Make sure you understand the course content. **See the syllabus for more details on the generative AI policy.**

Homework Assignments

- Real-world/public health/medical focus
- Scrape and wrangle/clean messy data
- Explore data
- Visualize data
- Perform statistical analyses
- Make predictions (build ML models)
- Communicate results
- Will be written in R using R Markdown and submitted via Canvas
- Must also submit knitted html file
- Points will be deducted if we are unable to knit your R Markdown file when grading
- Can use **2 late days per assignment**

Midterm Exam

- You will have 1 week to complete the midterm exam
- Very similar to a homework assignment
 - Similar length
 - Submit via Canvas
 - R Markdown and html files must be submitted
- No collaboration of any kind is permitted
- Use of generative AI is not permitted

Final Project

- Teams of 4-5 students
- Choose a project from list of 3
 - Will be given a dataset and prompt and will be tasked with data wrangling, data visualization, statistical analyses, building ML models and communicating your findings
- A TF will be assigned to each team to give advice throughout the project
 - Assigned in beginning of November

Course Materials

- 1. Lecture slides:** condensed content and main concepts
 - 2. R Markdown (.Rmd) files:** complete text explanations and example code shown in class, plus self-assessment questions and solutions and FAQs at the bottom
- All course material is available on Canvas organized by week and [GitHub](#) organized by topic (module)

Course Communication

- Email me or the teaching fellows
- Do **not** send a Canvas message – I rarely check them!
- Office hours
- Lab sessions
- See me during the class break or after lecture

Course Expectations

- You are **highly** encouraged, but not required, to attend lecture
 - Each lecture will be recorded and available on the Canvas course site
- Participation is not required or included as part of your final grade, but is highly encouraged
 - We all learn from each other
 - After a while I start to dislike the sound of my own voice
- Attending a weekly lab session is highly recommended but not required
 - The session will be recorded and available on the Canvas course site
- Break time - we'll take a 5- minute break around the middle of each lecture (45-50 minutes in)

Course Conduct: Chatham House Rule

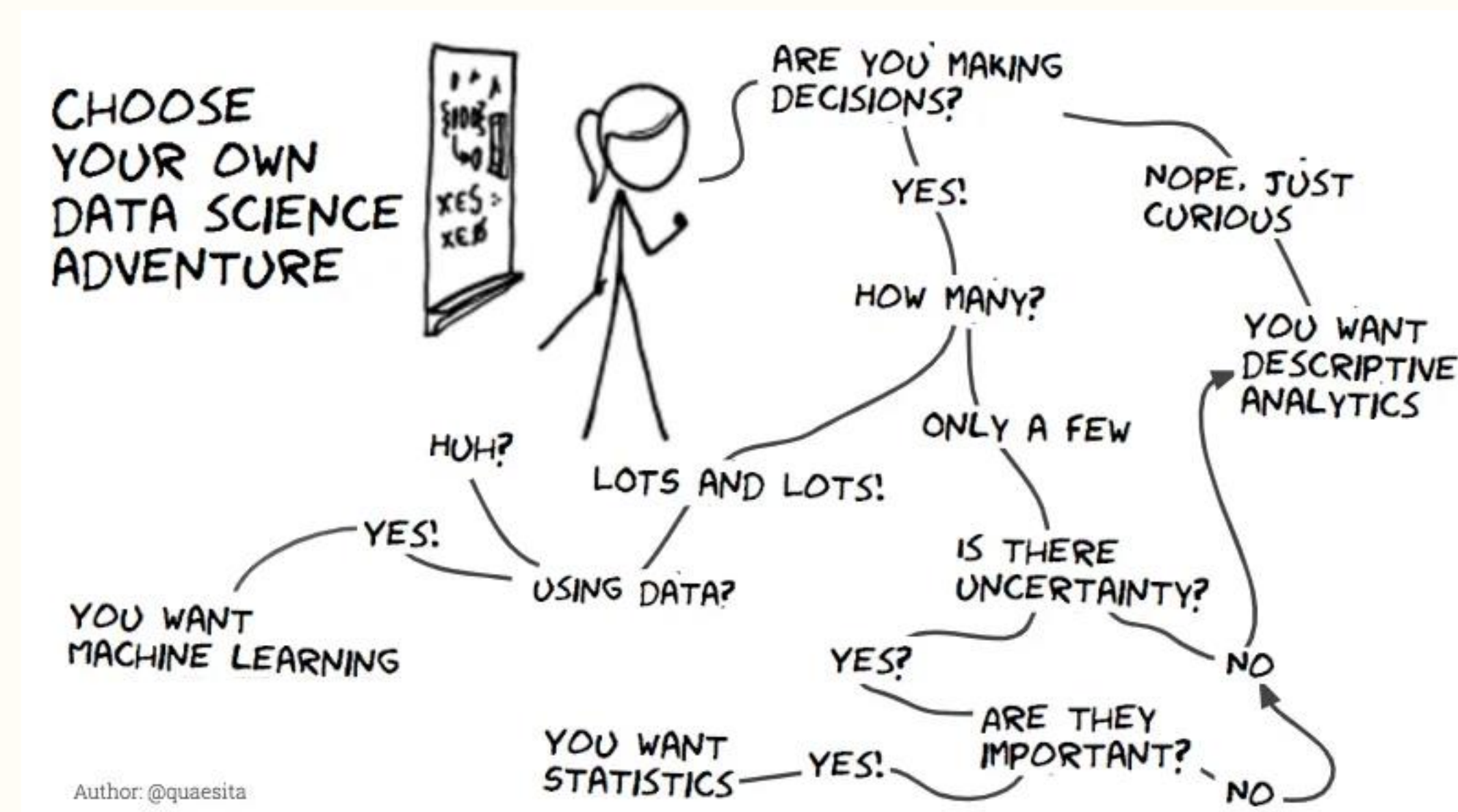
To encourage openness, trust and candid discussion across a variety of topics at Chan, I propose we adopt a **Non-Attribution Community Norm** for this course, modeled after the Chatham House Rule. This social contract means you may share the ideas discussed in class outside of it, but you may not identify who said them or what affiliation they hold.

This means:

- You are welcome to talk about class ideas outside of class.
- Do not quote classmates by name outside of class or on social media.
- Screenshots, recordings, or transcripts may **not** be circulated without written permission from faculty, speakers, etc.
- The rule also applies in breakout rooms (either online or in person).

This course IS...

- An introduction to R and RStudio
- An introduction to creating publishable visualizations using `ggplot2`
- An introduction to using git and GitHub for version control and reproducibility
- Focused on foundational knowledge for coding, the data science project pipeline, machine learning, and communication of process and results
- A chance for you to collaborate with and learn from other students
- A safe space to “get your hands dirty” and write and debug code



This course is NOT...

- A machine learning course
 - You will get a great foundation in ML from this course, but ML is only about 40% of the course
- Meant to be a silver bullet for all data science tasks / projects
 - Passing this course does not mean you won't have more to learn - data science is constantly evolving and we need to be lifelong learners

Action Items

- Download and install [R](#) and [RStudio](#)
 - Make sure you **download R first**
- Create a [GitHub account](#)
 - If you are interested in downloading course materials from GitHub
- Complete this [survey](#)

Introduction to R, RStudio, and R Markdown

Introduction to R

Heather Mattie, PhD

What is R?

R is a programming language and software environment specifically designed for **statistical computing** and **graphics**

- Open source
- Specialized for statistics
- Active community
- Reproducible research



[\[Home\]](#)

Download

[CRAN](#)

R Project

[About R](#)

[Logo](#)

[Contributors](#)

[What's New?](#)

[Reporting Bugs](#)

[Conferences](#)

[Search](#)

[Get Involved: Mailing Lists](#)

[Get Involved: Contributing](#)

[Developer Pages](#)

[R Blog](#)

The R Project for Statistical Computing

Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred [CRAN mirror](#).

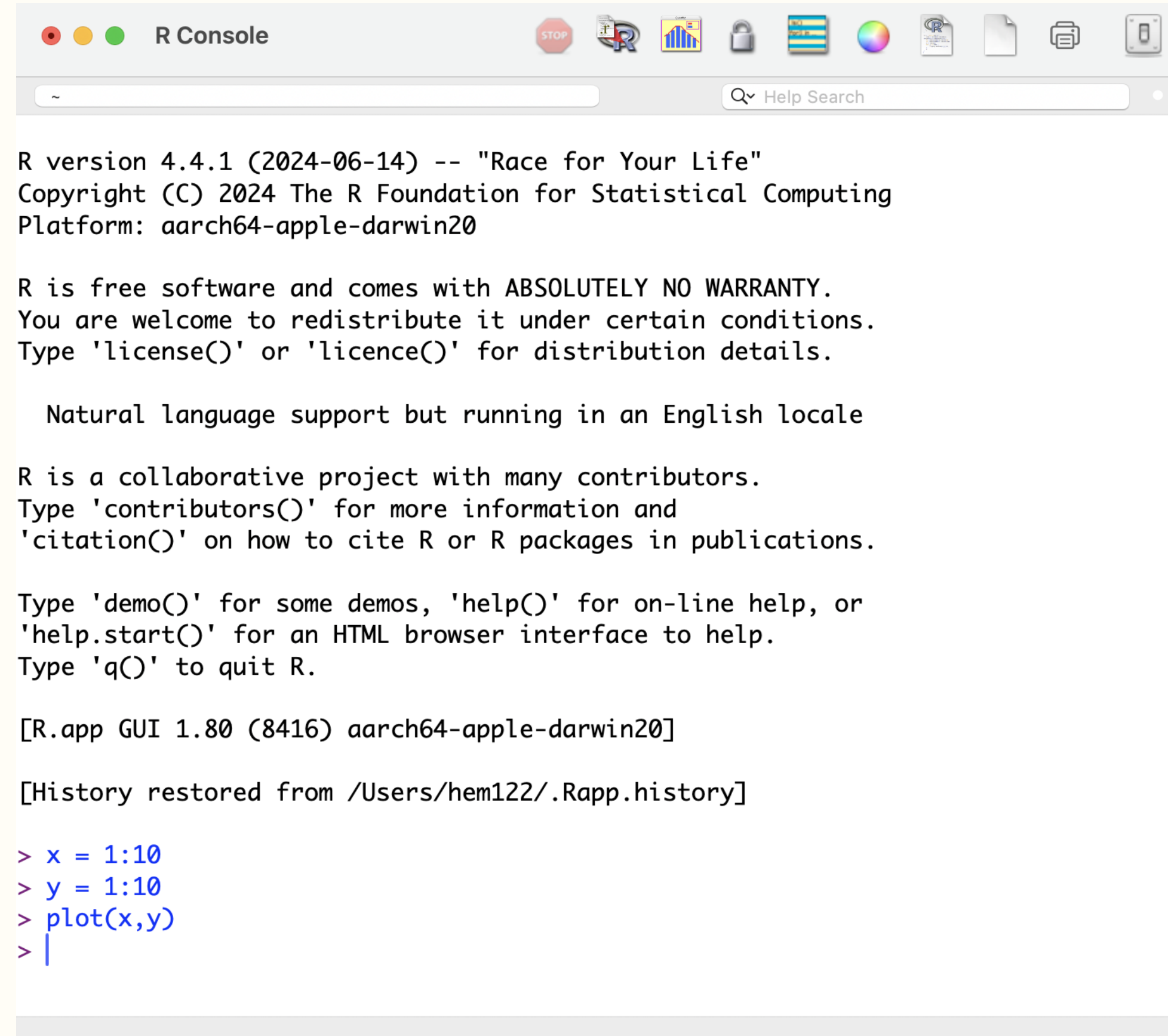
If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

News

- **R version 4.5.1 (Great Square Root)** has been released on 2025-06-13.
- **R version 4.5.0 (How About a Twenty-Six)** has been released on 2025-04-11.
- **R version 4.4.3 (Trophy Case)** (wrap-up of 4.4.x) was released on 2025-02-28.
- The **useR! 2025** conference will take place at Duke University, in Durham, NC, USA, August 8-10.
- We are deeply sorry to announce that our friend and colleague Friedrich (Fritz) Leisch has died. [Read our tribute to Fritz here.](#)



What is R?



```
R version 4.4.1 (2024-06-14) -- "Race for Your Life"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: aarch64-apple-darwin20

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

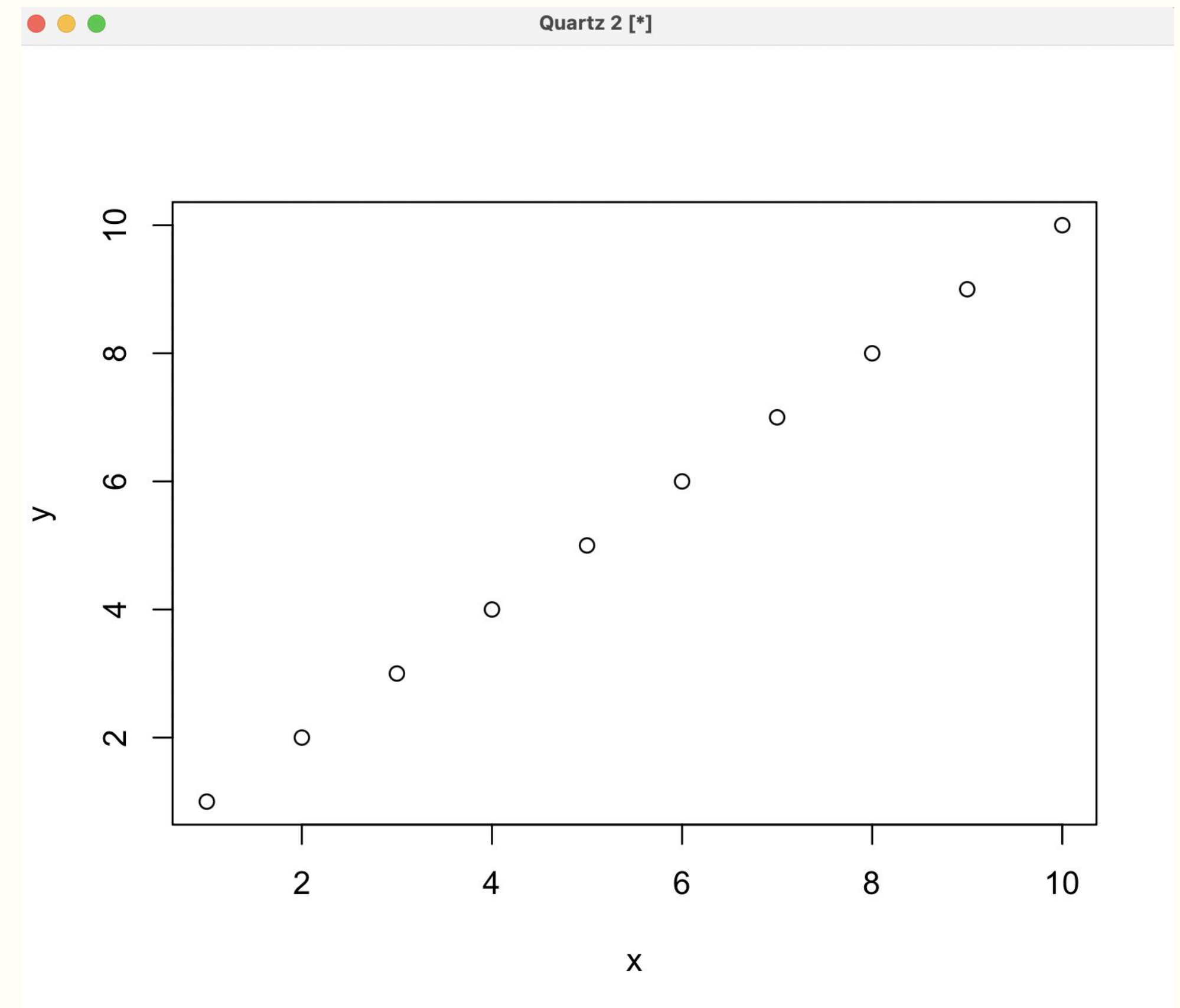
Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[R.app GUI 1.80 (8416) aarch64-apple-darwin20]
[History restored from /Users/hem122/.Rapp.history]

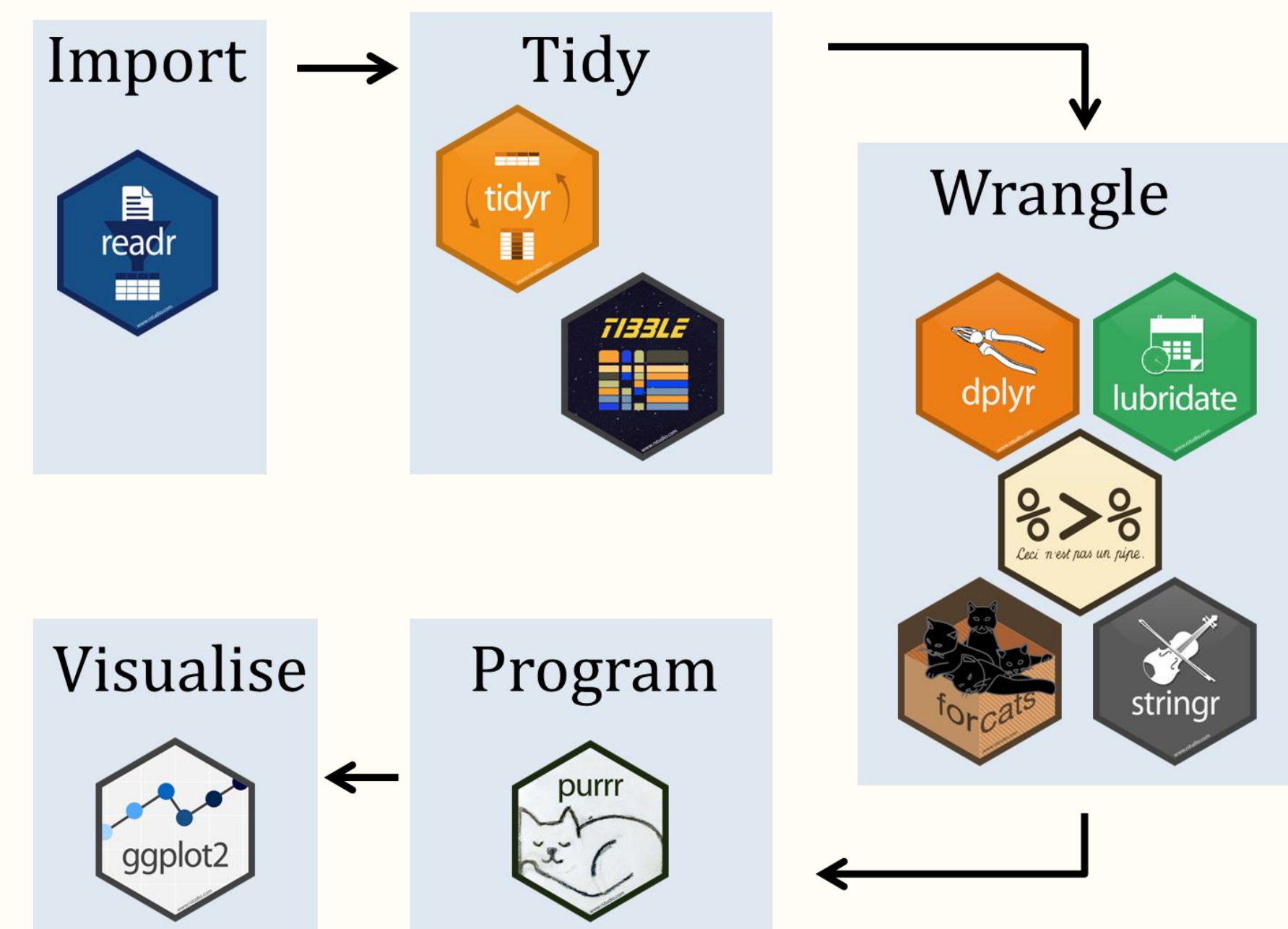
> x = 1:10
> y = 1:10
> plot(x,y)
> |
```



Packages

In R, **packages** are collections of functions, data, and documentation that extend the base functionality of the language.

- Allow users to perform specific tasks or analyze particular types of data
- Essentially bundles of reusable code that can be easily shared and installed
- Must first install a package, and then load it when you want to use it
- All have documentation available online



The R Community

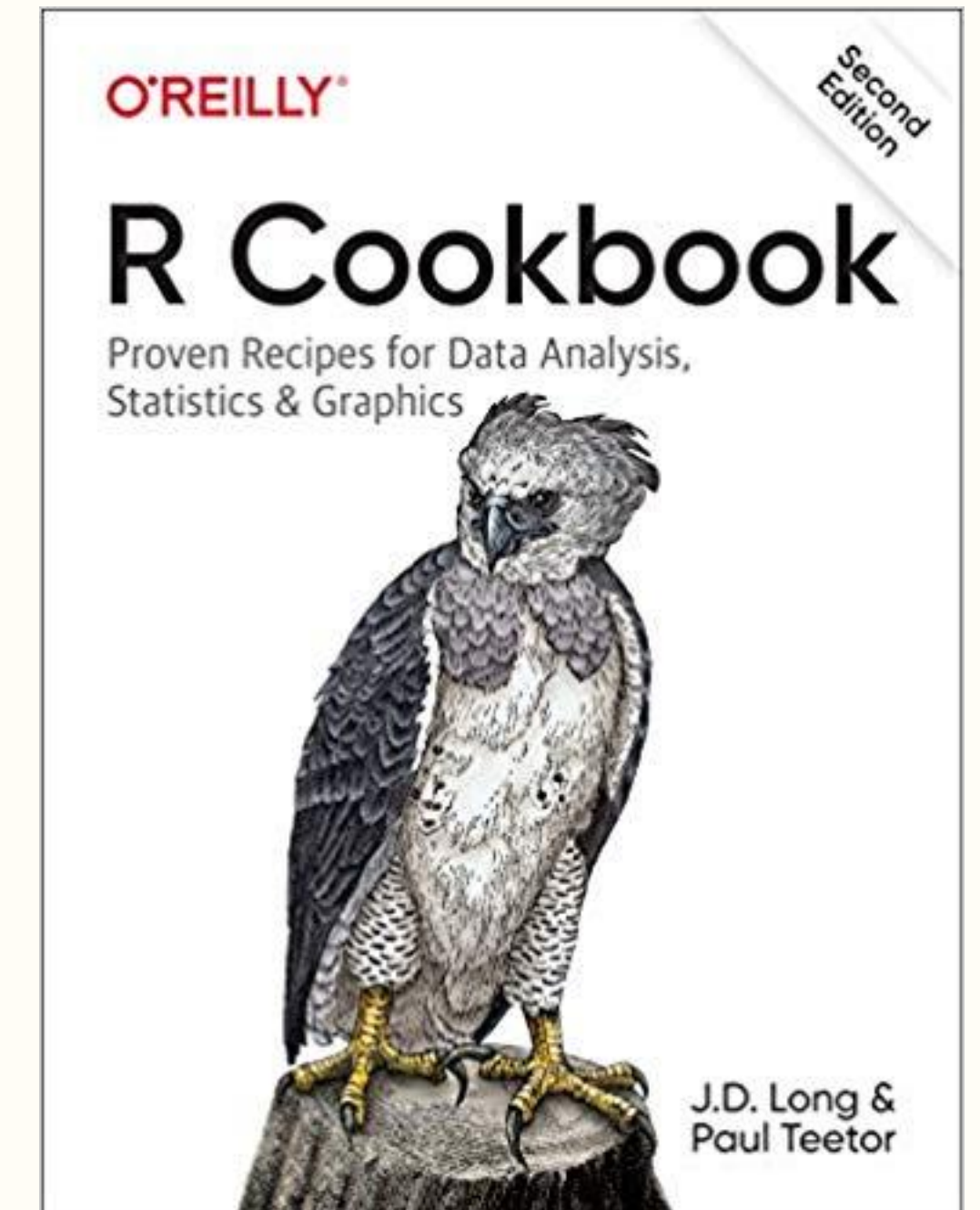
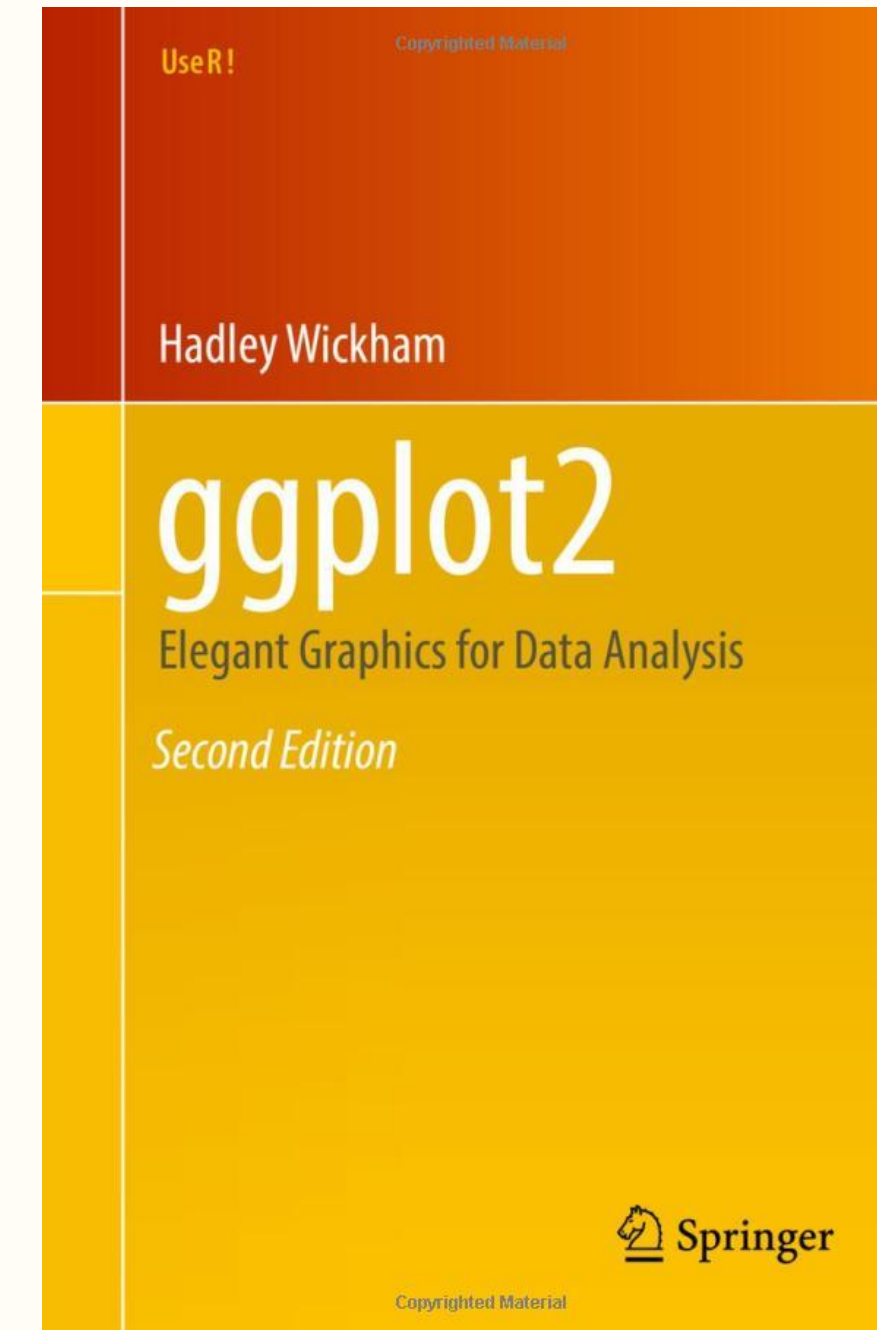
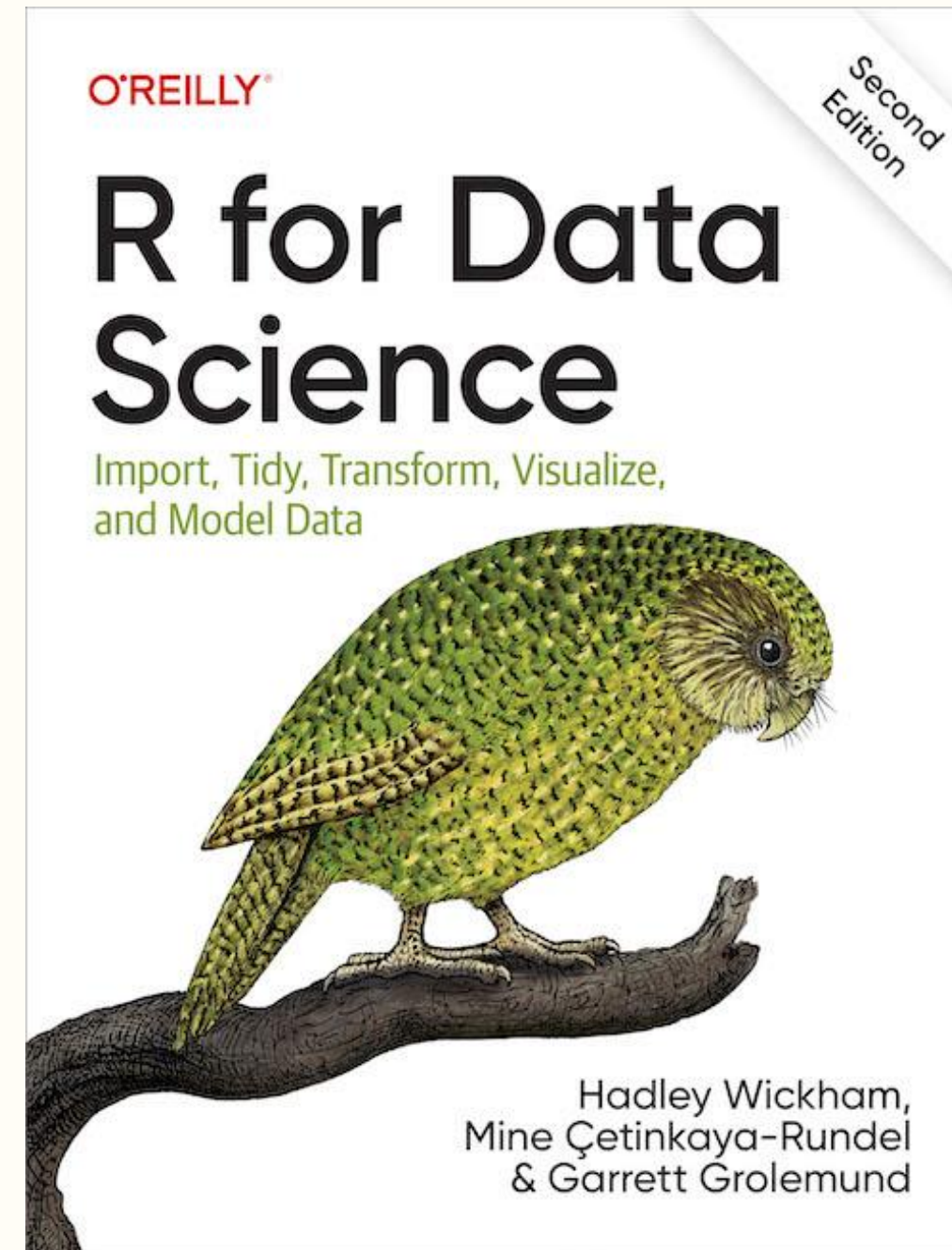
One of the best things about R is the **R community**. The R Community is a global network of individuals who use, contribute to, and promote the R programming language.

- Active engagement through online forums, blogs, social media, and conferences
- Members range from beginners to experts in various fields
- The community fosters collaboration, knowledge sharing, and support for users of all levels

R Resources

For help writing code:

- R for Data Science
- ggplot2
- R Cookbook



R Resources

For help **debugging** code:

- R Help: `help()` and ?
- Vignettes and code demonstrations
- Package documentation
- Stack Overflow

stackoverflow About Products OverflowAI [r]

Home Questions Tags Challenges NEW Chat Articles Users Jobs Companies

COLLECTIVES + Communities for your favorite technologies. [Explore all Collectives](#)

[r]

R is a free, open-source programming language and software environment for statistical computing, bioinformatics, information graphics, and general computing.

Sign up to watch this tag and see more personalized content

Watch tag Go to Wiki

510,956 questions Newest Active Bountied Unanswered More Filter

1 vote 0 answers 22 views Method to set minimum column width in flextable

I'm looking for a way to apply `autofit()` only to columns that are too small. This is my current method but I'm looking for a better way to do it: `library(flextable) library(magrittr) ### Ex. Tbls ...`

R r r-flextable Aaron Rose 11 asked 3 hours ago

-1 votes Extract last numbers from text strings [duplicate]

ie with a column of strings that have either 1 or 2 numbers at the end of the
:t these 1 or 2 numbers and place them in a new column. Here is how I am ...

jr Eric Krantz 2,257 asked 3 hours ago

age related to geocoding that could help me assign cate-
closed]

project where I am trying to compile polygons of various Parks at different
national, etc.) In my dataset, I only have data on the name of each Park, an...

ing reverse-geocoding sam 1 asked 6 hours ago

Package ‘dslabs’

March 1, 2024

Title Data Science Labs

Version 0.8.0

Description Datasets and functions that can be used for data analysis practice, home-work and projects in data science courses and workshops. 26 datasets are available for case studies in data visualization, statistical inference, modeling, linear regression, data wrangling and machine learning.

Author Rafael A. Irizarry, Amy Gill

Maintainer Rafael A. Irizarry <rafael_irizarry@dfci.harvard.edu>



Summary

- R is a programming language designed for statistical computing and visualizations
- R is open source and has a large, active, and diverse community that contributes to and promotes R
- R packages are collections of functions, data, and documentation that extend the base functionality of R
- There are many resources for writing and debugging R code

Introduction to R, RStudio, and R Markdown

Introduction to RStudio

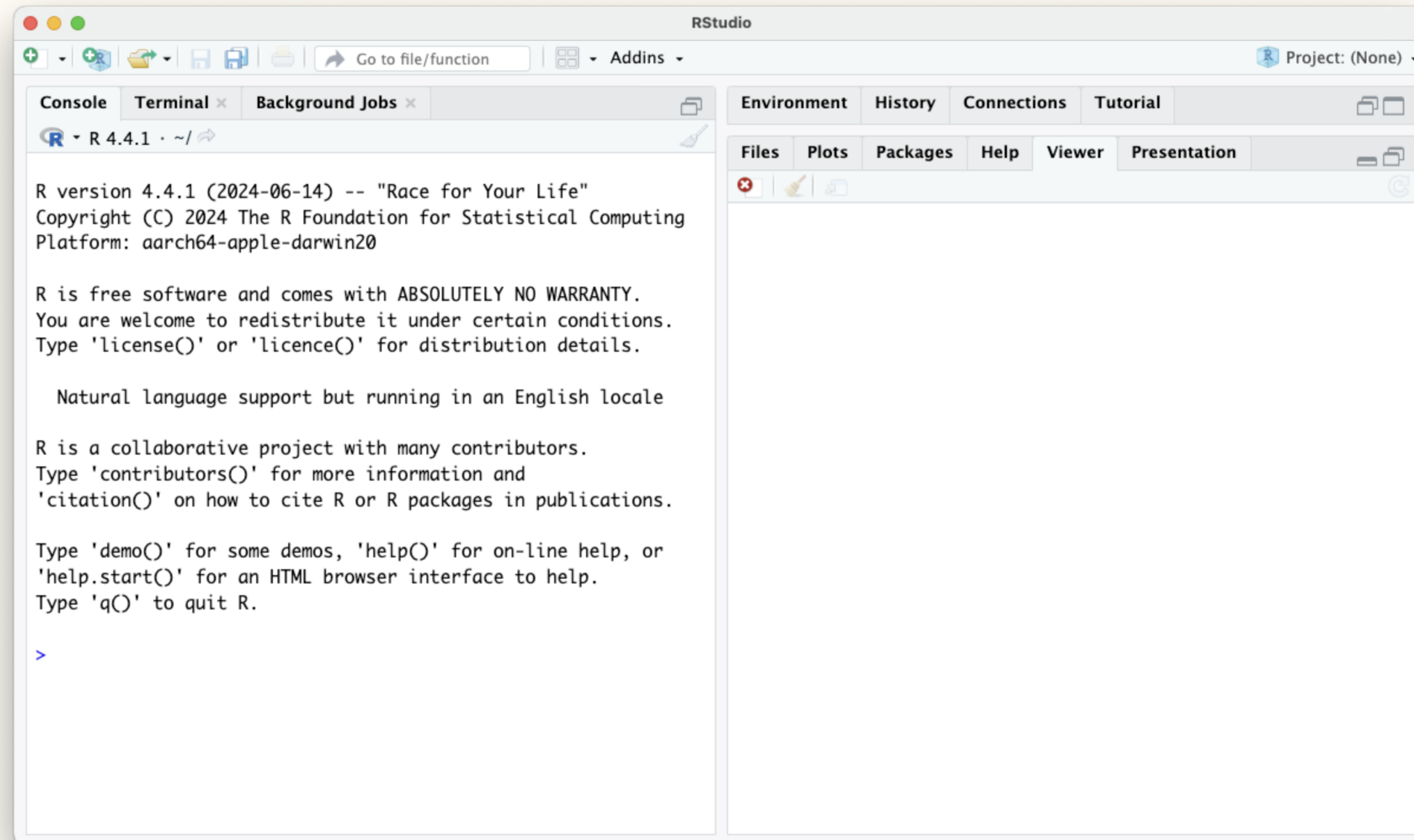
Heather Mattie, PhD

What is RStudio?

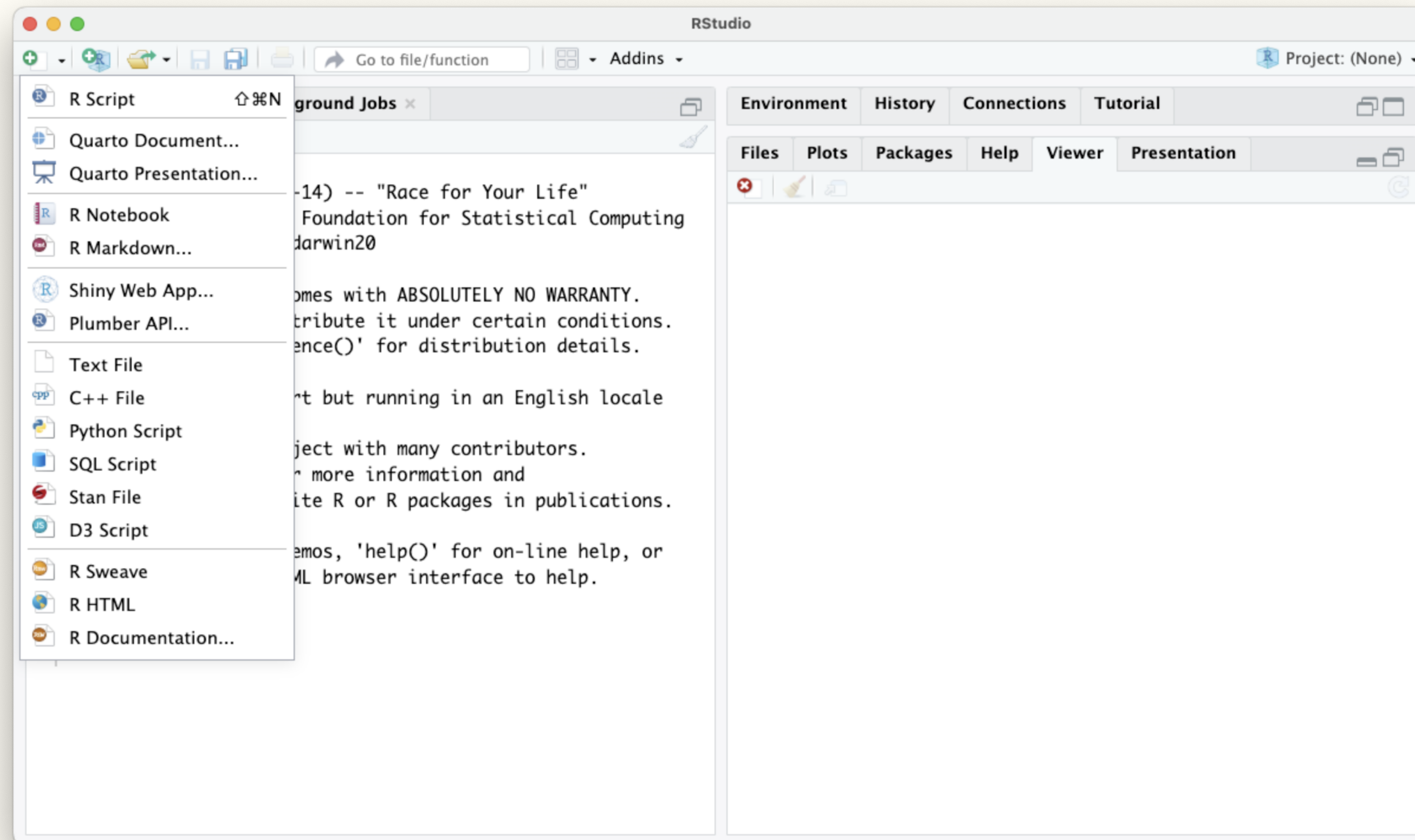
- **RStudio** is an integrated development environment (IDE) specifically created for R
 - RStudio desktop
 - User-friendly interface



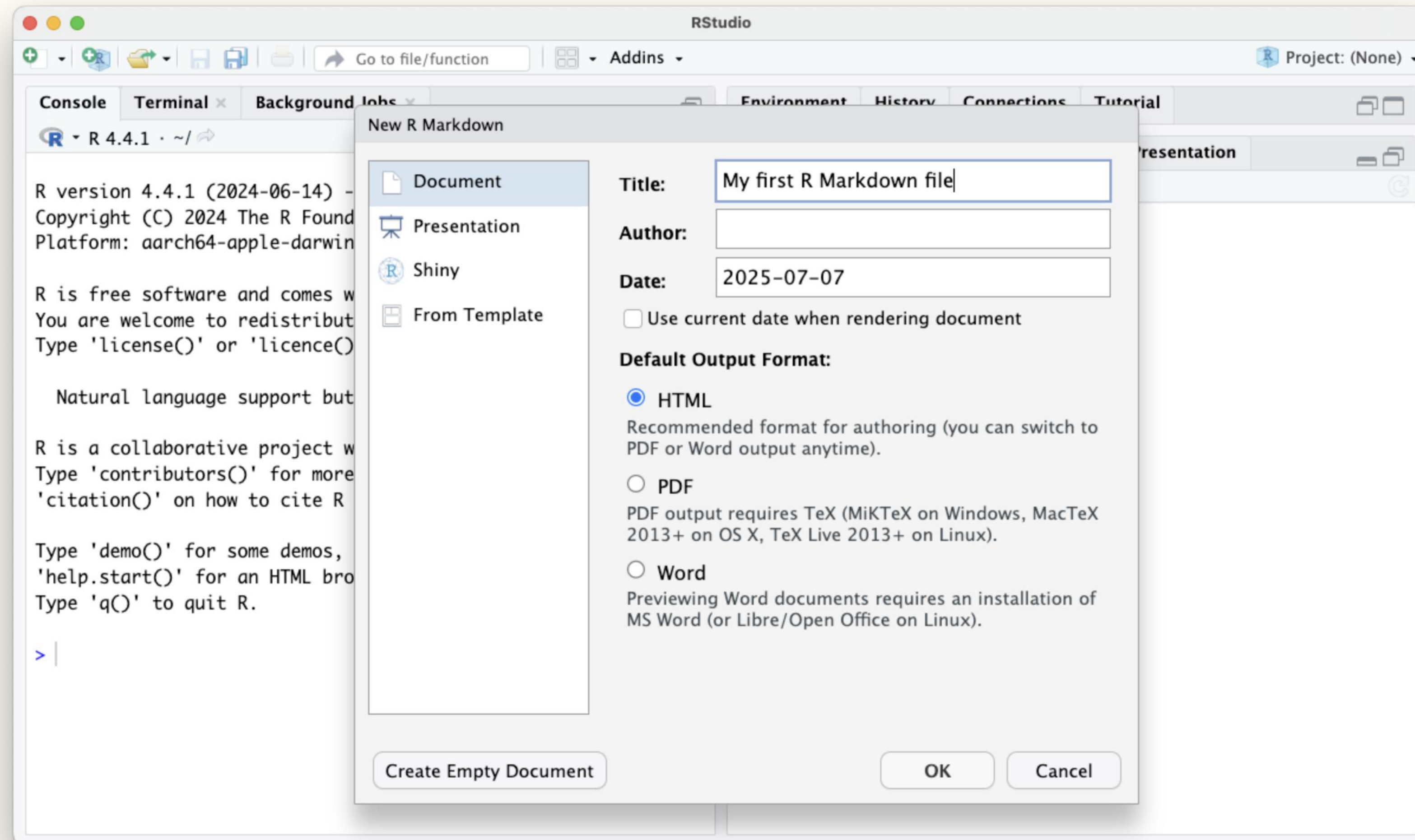
Layout



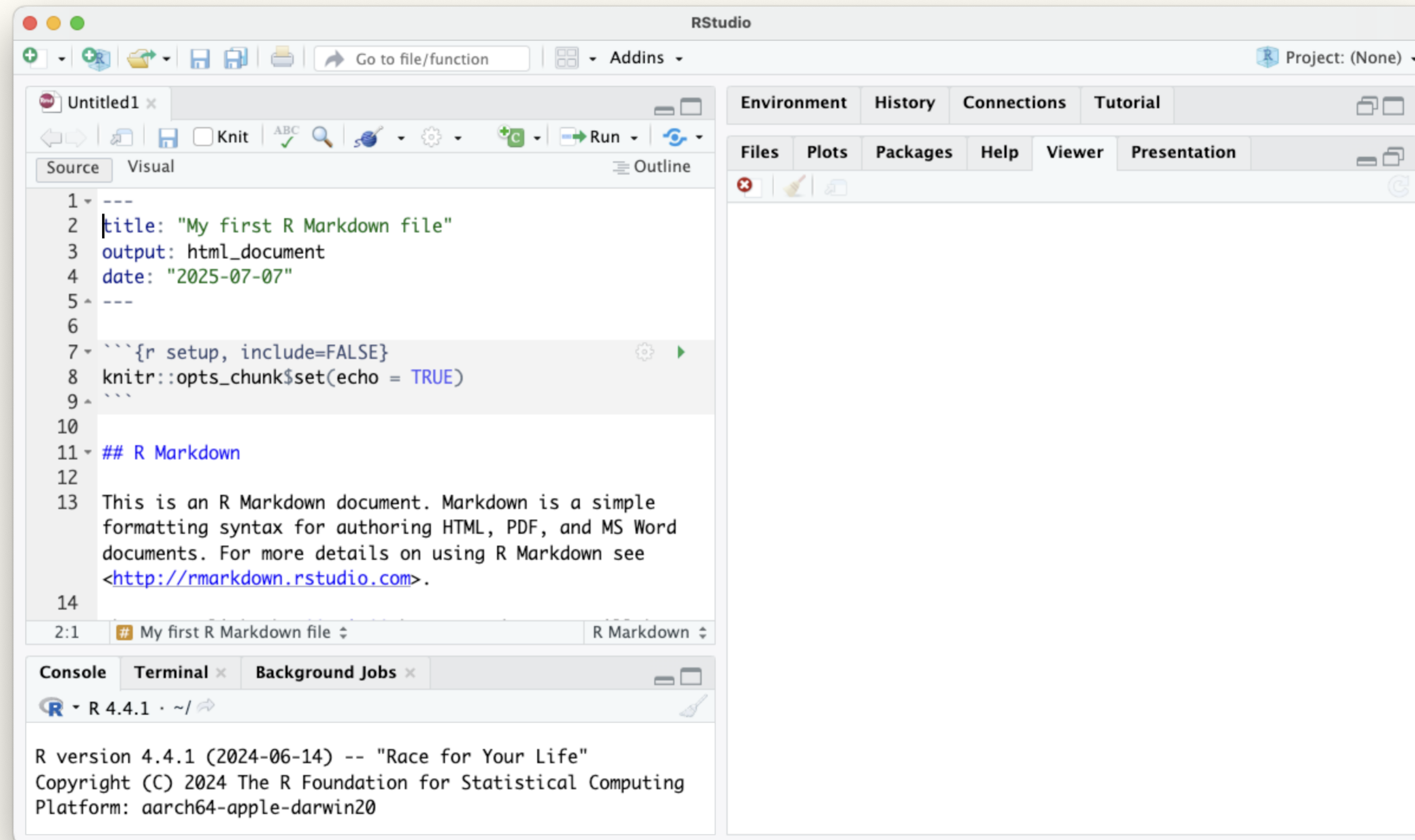
Layout



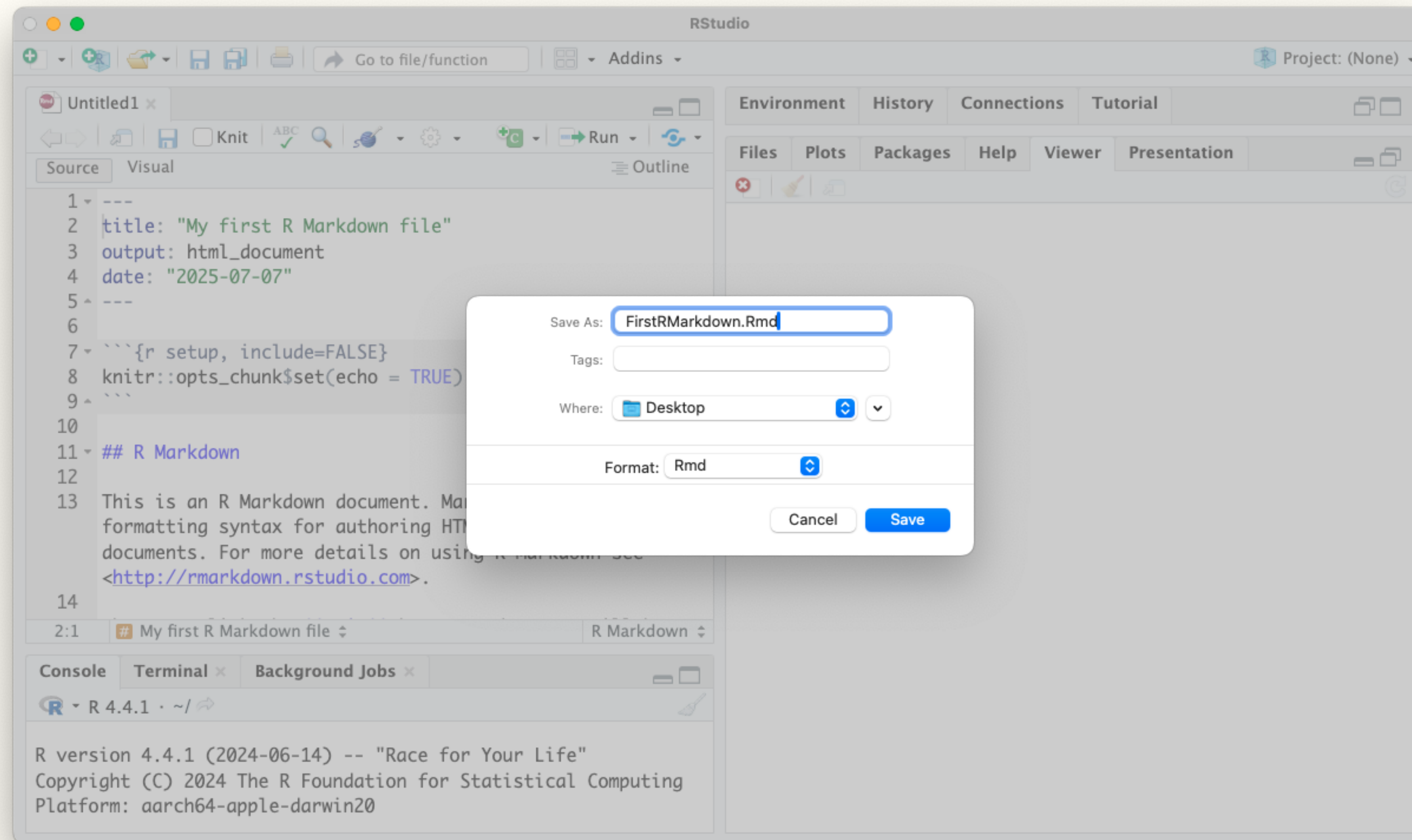
Layout



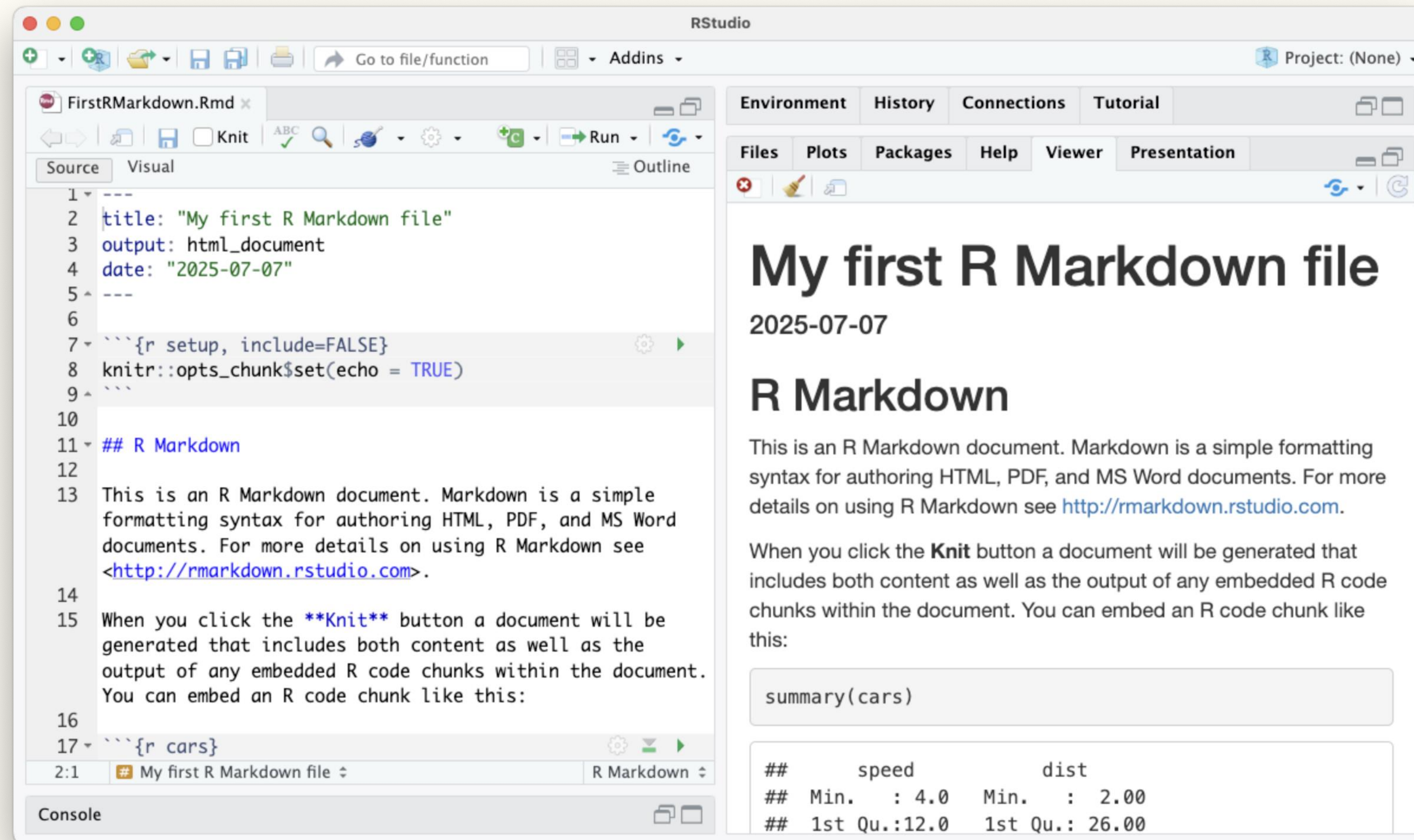
Layout



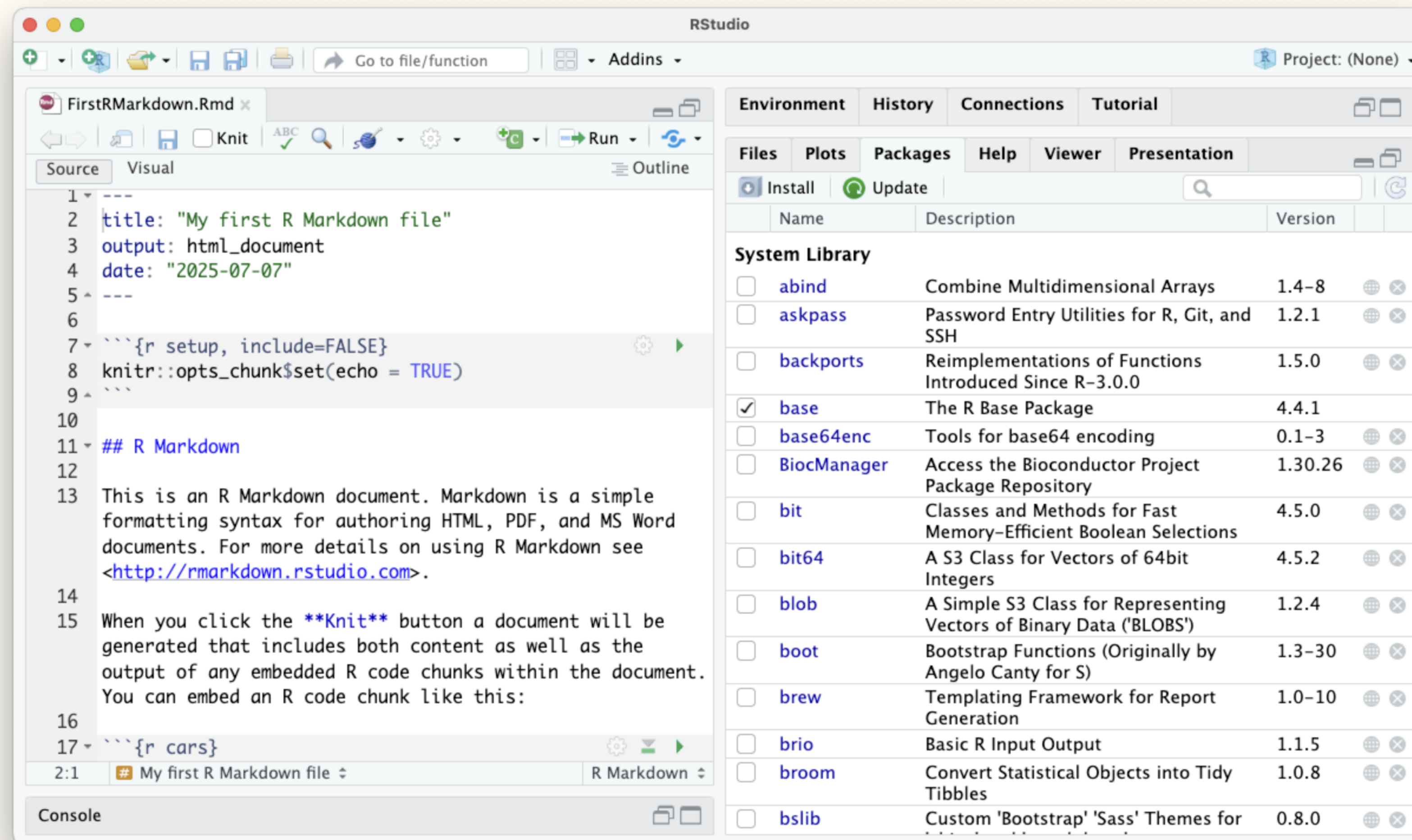
Layout



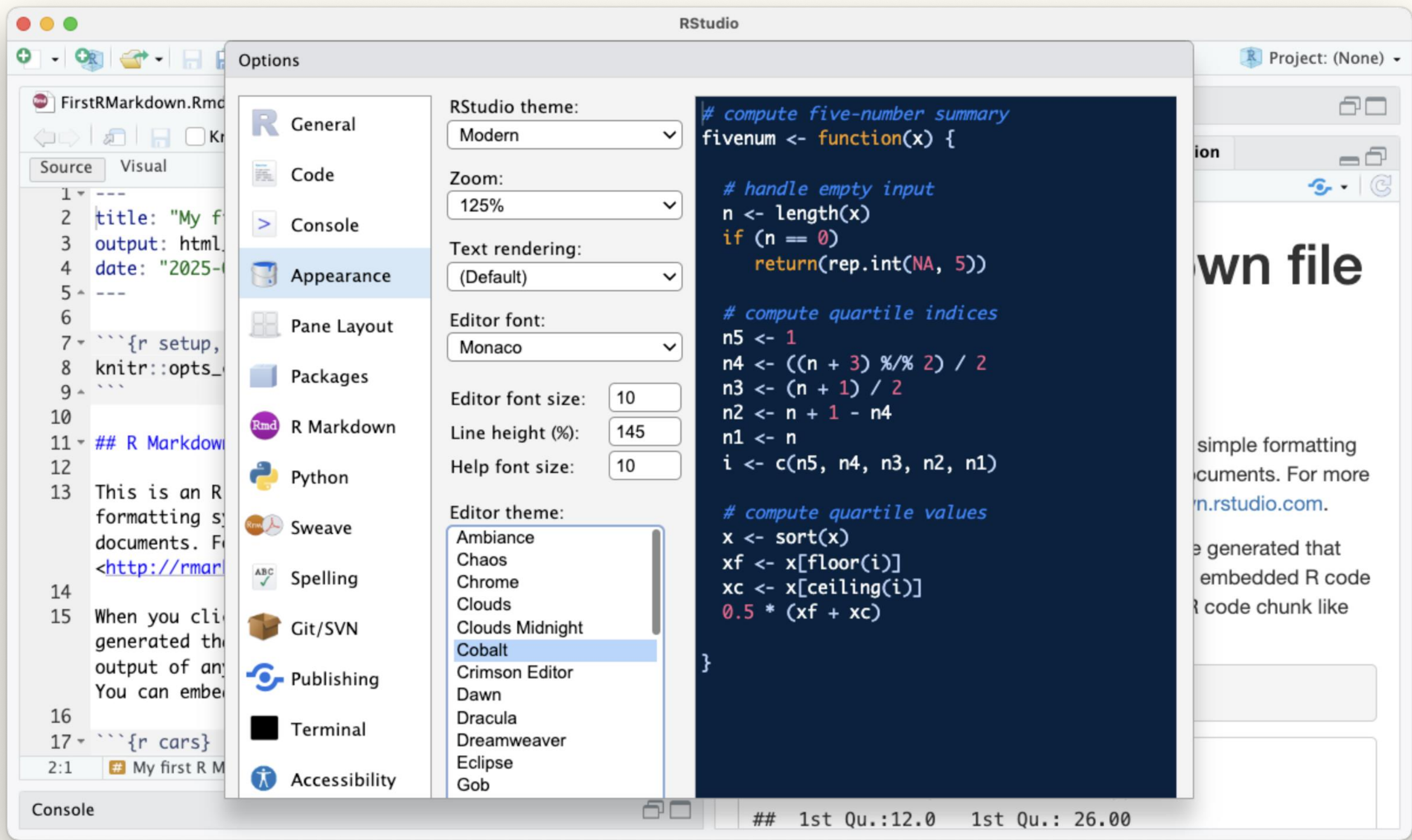
Layout



Layout



Appearance



Summary

- RStudio is an integrated development environment specifically created for R
- RStudio provides a user-friendly interface
- RStudio windowpanes showcase code, an opened file, output, etc.
- The appearance of RStudio can be changed to your preferred style

Introduction to R, RStudio and R Markdown

Introduction to R Markdown

Heather Mattie, PhD

What is R Markdown?

- **R Markdown** is a tool for creating dynamic documents that **combine** text, code, and the output of that code into a **single** file



SOURCE EDITOR

1. New File

5. Save and Render

6. Share

4. Set Output Format(s) and Options

3. Write Text

2. Embed Code

```
1 ---
2 title: "Document Title"
3 author: "Author Name"
4 output:
5   html_document:
6     toc: TRUE
7 ---
8
9 ```{r setup, include=FALSE}
10 knitr::opts_chunk$set(echo = TRUE)
11 ```
12
13 ## R Markdown
14
15 This is an R Markdown document.
16 Markdown is a simple formatting
17 syntax for authoring HTML, PDF,
18 and MS Word documents.
19
20 ```{r cars}
21 summary(cars)
22 ```
```

set preview location

insert code chunk

go to code chunk

run code chunk(s)

show outline

run all previous chunks

modify chunk options

run current chunk

Console Terminal R Markdown Jobs

~/Desktop/report/report.Rmd

Output created: report.html

RENDERED OUTPUT

file path to output document

report.html Open in Browser Find Publish

find in document

Document Title

Author Name

- R Markdown
- Including Plots

R Markdown

reload document

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

summary(cars)

##	speed	dist
## Min.	: 4.0	Min. : 2.00
## 1st Qu.:	12.0	1st Qu.: 26.00
## Median :	15.0	Median : 36.00
## Mean :	15.4	Mean : 42.98
## 3rd Qu.:	19.0	3rd Qu.: 56.00
## Max. :	25.0	Max. : 120.00

report.html 802.9 KB Jul 9, 2021, 4:35 PM

What is Markdown?

Markdown is a lightweight markup language that uses simple text formatting to structure and style text.

- Plain text formatting
- Easy formatting
- Readability
- Versatile

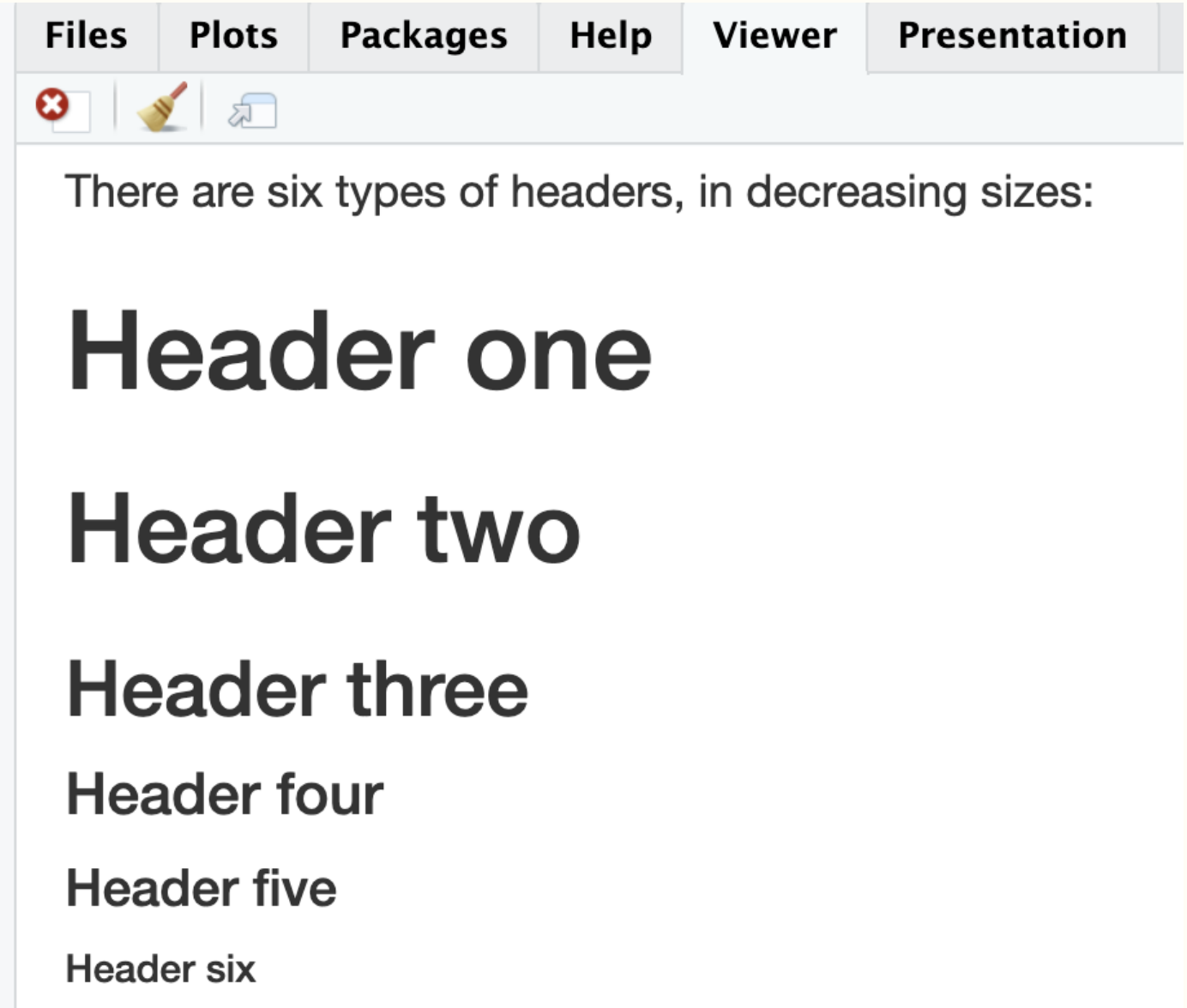


Headings

To denote a **heading**, add a **number sign** (hashtag) before it

- The more number signs, the smaller the text

```
43
44
45
46 There are six types of headers, in decreasing
    sizes:
47
48 ▾ # Header one
49
50 ▾ ## Header two
51
52 ▾ ### Header three
53
54 ▾ #### Header four
55
56 ▾ ##### Header five
57
58 ▾ ##### Header six
59
60
```



Bold and Italics

To make a word or string of words *italic*, surround the word or words with 1 asterisk (*) or 1 underscore (_)

To make a word or string of words **bold**, surround the word or words with 2 asterisks (**) or 2 underscores (__)

```
29 ▾ ### Italics and Bold
30
31 _This_ is one word in italics.
32
33 _This is a string of words in italics._
34
35 **This** is one word in bold.
36
37 **This is a string of words in bold.**
38
39 _**This is a few words**_ in bold and
    _italics_.
```

Italics and Bold

This is one word in italics.

This is a string of words in italics.

This is one word in bold.

This is a string of words in bold.

This is a few words in **bold** and *italics*.

Lists

You can create **unordered** and **ordered** lists in Markdown.

- To create an **unordered** list, preface each item with an asterisk (*) and give each list item its own line.
- To create an **ordered** list, preface each item with a number and give each item its own line.

139 To create an unordered list, you'll want to preface each item in the
140 list with an asterisk (`*``). Each list item also gets its own line. For
141 example, a grocery list in Markdown might look like this:

```
141 * Milk  
142 * Eggs  
143 * Salmon  
144 * Butter
```

145
146 An ordered list is prefaced with numbers, instead of asterisks. For
example:

```
147  
148 1. Crack three eggs over a bowl  
149 2. Pour a gallon of milk into the bowl  
150 3. Rub the salmon vigorously with butter  
151 4. Drop the salmon into the egg-milk bowl
```

To create an unordered list, you'll want to preface each item in the list with an asterisk (*). Each list item also gets its own line. For example, a grocery list in Markdown might look like this:

- Milk
- Eggs
- Salmon
- Butter

An ordered list is prefaced with numbers, instead of asterisks. For example:

1. Crack three eggs over a bowl
2. Pour a gallon of milk into the bowl
3. Rub the salmon vigorously with butter
4. Drop the salmon into the egg-milk bowl

Hyperlinks

There are two link types in Markdown:

1. To create an **inline link**, wrap the link text in brackets ([]) and the link in parentheses ().
2. A **reference link** is used when you want to reference the same link multiple times in your document. A specific name is selected as the reference, put in square brackets, and then a colon and the link are placed next to it. To use the reference link, wrap the link text in brackets and the reference link name in brackets.

```
90 ▾ ### Links
91
92 ▾ #### Inline links
93
94 Visit [this website](https://rmarkdown.rstudio.com/lesson-1
    .html) for more details about R Markdown!
95
96
97 ▾ #### Reference links
98
99 Here's [a link to something else][another place].
100 Here's [yet another link][another link].
101 And now back to [the first link][another place].
102
103 [another place]: https://rstudio.github.io/cheatsheets/
104 [another link]: https://www.hsph.harvard.edu/
```

Links

Inline links

Visit [this website](https://rmarkdown.rstudio.com/lesson-1.html) for more details about R Markdown!

Reference links

Here's [a link to something else](#).

Here's [yet another link](#).

And now back to [the first link](#).

Images

There are two image types in Markdown:

1. To create an **inline image**, use the exact same syntax as an inline hyperlink, but with an exclamation point (!) in the beginning.
2. A **reference image** also uses nearly the same syntax as a reference hyperlink, but with an exclamation point (!) in the beginning.

Images

There are two image types in Markdown:

1. To create an **inline image**, use the exact same syntax as an inline hyperlink, but with an exclamation point (!) in the beginning.
2. A **reference image** also uses nearly the same syntax as a reference hyperlink, but with an exclamation point (!) in the beginning.

```
125 ▾ ### Images
126
127 ▾ #### Inline images
128
129 For example, to create an inline image link to
130 https://octodex.github.com/images/bannekat.png, with an alt text
131 that says, Benjamin Bannekat, you'd write this in Markdown:
132
133 !\[Benjamin Bannekat\](https://octodex.github.com/images/bannekat.png)
134 { width=25% }
135
```

Images

Inline images

For example, to create an inline image link to <https://octodex.github.com/images/bannekat.png>, with an alt text that says, Benjamin Bannekat, you'd write this in Markdown:



Benjamin Bannekat

Images

There are two image types in Markdown:

1. To create an **inline image**, use the exact same syntax as an inline hyperlink, but with an exclamation point (!) in the beginning.
2. A **reference image** also uses nearly the same syntax as a reference hyperlink, but with an exclamation point (!) in the beginning.

```
136 - #### Reference images
137
138 ![The first father][First Father]
139
140 ![The second first father][Second Father]
141
142 [First Father]: http://octodex.github.com/images/founding-father.jpg { width=50% }
143
144 [Second Father]:http://octodex.github.com/images/foundingfather\_v2.png { width=50% }
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```

Reference images



The first father

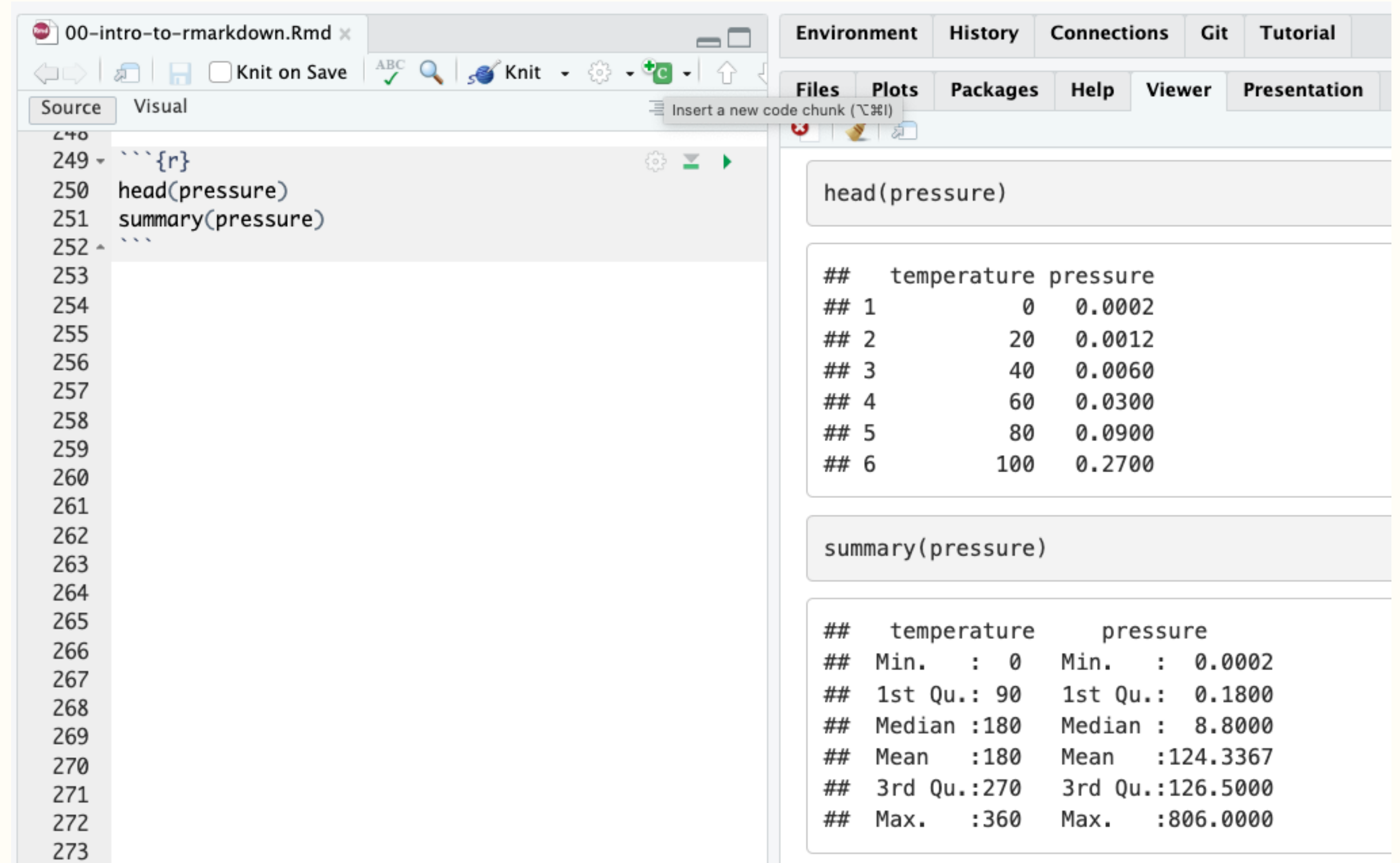


The second first father

Embedding code

There are three ways to create a code chunk:

1. Typing ``{r}`` to start a code chunk and then ``` to end it
2. Clicking the **Add Chunk** button in the tool bar
3. The keyboard shortcut
Ctrl + Alt + I
(Mac: **Cmd + Option + I**)



The screenshot shows the RStudio interface with a file named "00-intro-to-rmarkdown.Rmd". The "Source" pane on the left shows a code chunk starting with ``{r}`` and containing the following R code:

```
head(pressure)
summary(pressure)
```

The "Environment" pane on the right shows the output of the code chunk. It displays the results of `head(pressure)` and `summary(pressure)`.

head(pressure)

##	temperature	pressure
## 1	0	0.0002
## 2	20	0.0012
## 3	40	0.0060
## 4	60	0.0300
## 5	80	0.0900
## 6	100	0.2700

summary(pressure)

##	temperature	pressure
## Min. :	0	Min. : 0.0002
## 1st Qu.: 90		1st Qu.: 0.1800
## Median :180		Median : 8.8000
## Mean :180		Mean :124.3367
## 3rd Qu.:270		3rd Qu.:126.5000
## Max. :360		Max. :806.0000

Embedding code

There are different display options for code and code output

- Adding a comma and **echo = FALSE** after the **r** displays the code output but not the code itself
- Adding a comma and **include = FALSE** prevents code and output from appearing in the finished file

```
249 ```{r, echo = FALSE}
250 head(pressure)
251 summary(pressure)
252 ```
253
254
255
256
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258
259
260
261
262
263
264
265
266
267
268
```

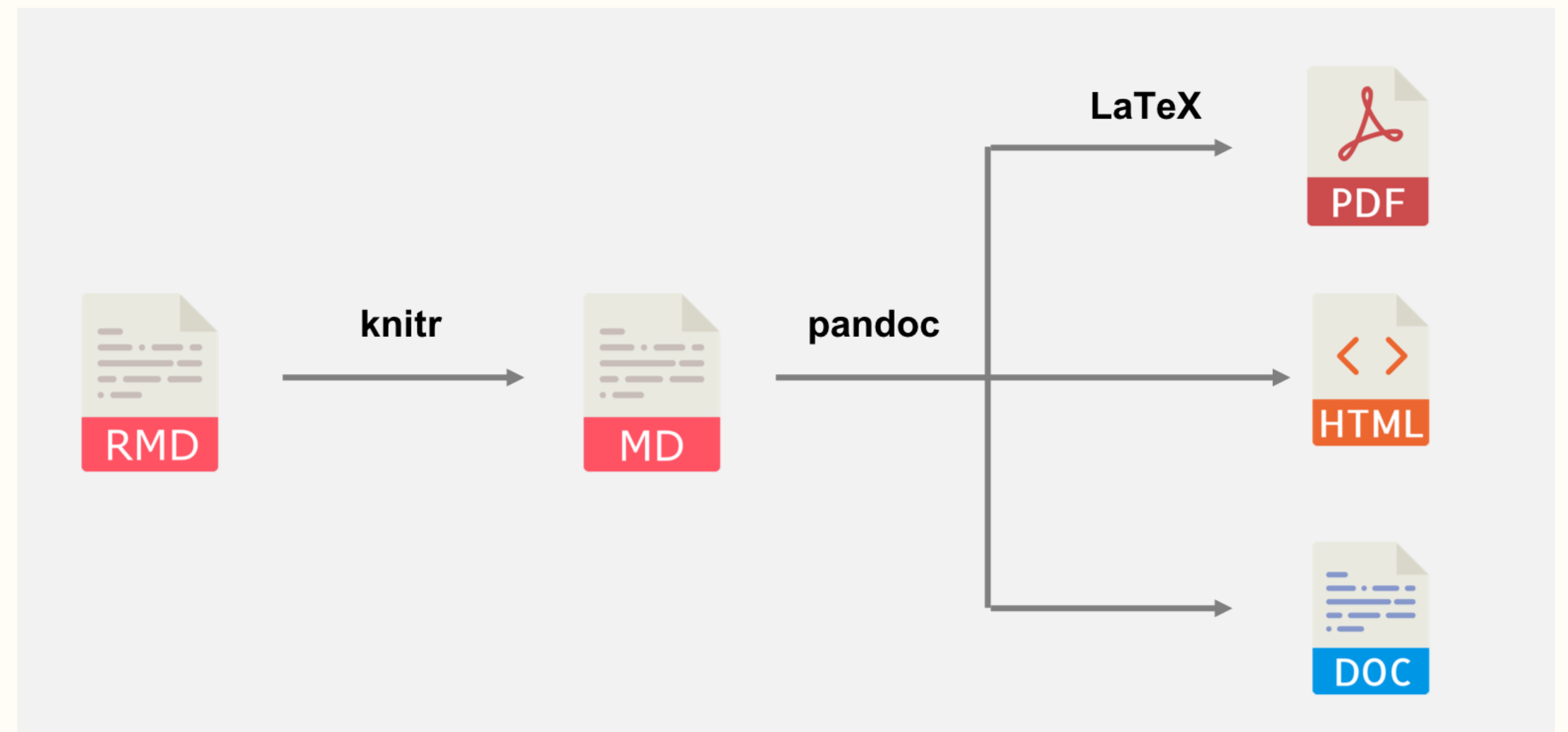
##	temperature	pressure
## 1	0	0.0002
## 2	20	0.0012
## 3	40	0.0060
## 4	60	0.0300
## 5	80	0.0900
## 6	100	0.2700

##	temperature	pressure
## Min. :	0	Min. : 0.0002
## 1st Qu.: 90		1st Qu.: 0.1800
## Median :180		Median : 8.8000
## Mean :180		Mean :124.3367
## 3rd Qu.:270		3rd Qu.:126.5000
## Max. :360		Max. :806.0000

Compiling R Markdown files

How does an R Markdown file (.Rmd file) compile?

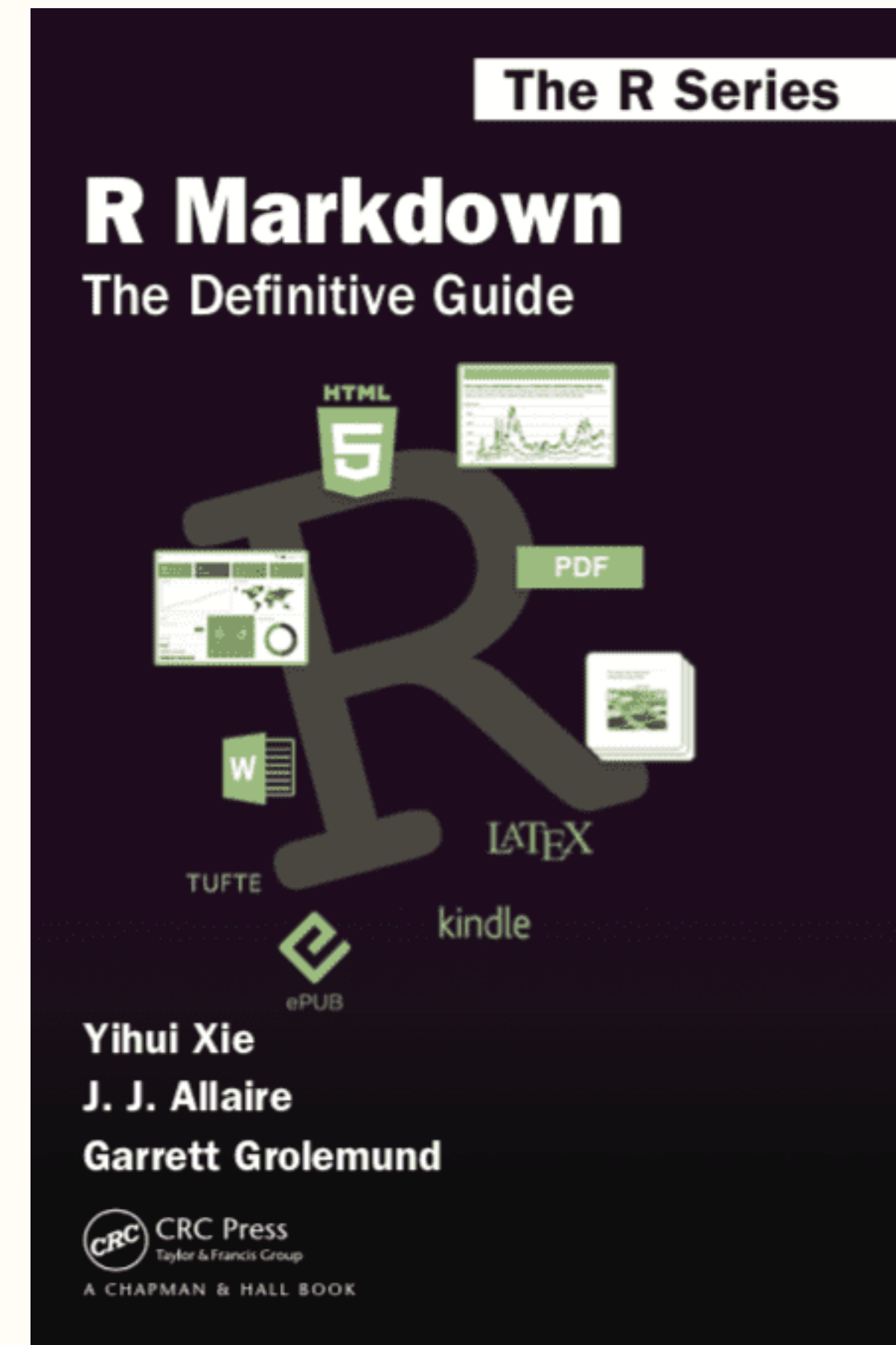
- The **rmarkdown** package renders an R Markdown file by executing other packages that knit the text and code chunks together into an output file
- The output file can be a Word doc, PDF, or HTML file
- The file will not knit (compile) if there is a bug in the code



Other document types

R Markdown can also be used to produce

- PDF presentations with beamer
- HTML presentations with ioslides and slidy
- Full academic manuscripts
- Dashboards
- Websites
- Interactive documents



Summary

- R Markdown is a tool that can create documents, slides, websites, etc. that combine text and code into one file
- Markdown is the text formatting markup language used in R Markdown files